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Latent information-guided one-step multi-view fuzzy clustering based on cross-view anchor graph

Chuanbin Zhang [a,b], Long Chen [a,*], Zhaoyin Shi [a], Weiping Ding [c]

[a] Department of Computer and Information Science, Faculty of Science and Technology, University of Macau, 999078, Macao Special Administrative Region of China
[b] School of Computer Science and Software, Zhaoqing University, Zhaoqing, 526061, China
[c] School of Information Science and Technology, Nantong University, Nantong, 226019, China

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ABSTRACT

Although graph-inspired clustering methods have achieved impressive success in the area of multi-view data analysis, current methods still face several challenges. First, classical graph construction approaches are computationally expensive in terms of both time and space for large-scale data. Second, fusing graphs from different views remains a challenge. Third, most existing methods require additional post-processing steps to generate the label assignment matrix. To address the above challenges, this paper proposes a novel multi-view clustering algorithm, called latent information-guided one-step multi-view fuzzy clustering based on cross-view anchor graph. Specifically, we introduce an efficient cross-view anchor graph learning approach to construct the similarity matrix of multi-view data and extract latent information from the graph, which reduces the computational complexity of graph learning and inspires the optimization of the consensus membership matrix. Following that, our one-step multi-view fuzzy clustering algorithm can directly generate the final clustering result in an effective and straightforward manner. Furthermore, during the optimization process, the proposed method balances consensus matrix learning and view-specified membership exploration via a self-tuned weighting mechanism. The comprehensive experimental analysis demonstrates the superiority of our approach over the state-of-the-art approaches.

1. Introduction

Multi-view data refers to a collection of information that comprises multiple features or modalities associated with the same set of objects. Examples of multi-view data include images captured from different perspectives of an object and news articles written in several languages. Given the wide prevalence of multi-view data in real-world tasks, multi-view data analysis emerges as a current research hotspot [1–3]. Multi-view clustering, being an unsupervised learning method, has garnered considerable attention from researchers [4–7]. Various approaches have been investigated in this field, such as spectral clustering [8–11], graph-based methods [12–14], subspace learning [15–17], K-means-based methods [18–20], and fuzzy clustering [21–23], etc.

Graph learning is naturally integrated into the multi-view clustering methods that rely on subspace learning and spectral clustering techniques [12–14,24]. Spectral clustering involves decomposing the data's Laplacian matrix to compute eigenvectors. On the other hand, subspace learning, especially the self-expression-based model, focuses on learning the similarity matrix of the target features. Kang et al. [15] utilized self-expression learning to obtain view-specific similarity matrices and proposed a structured graph learning approach that performs spectral clustering and graph fusion simultaneously. The paper [16] introduced a dual learning framework that learns the consensus self-expression matrix of both view-shared and view-specified features from the input data. Zhang et al. [17] explored the subspace representation of latent features underlying data, resulting in a robust and precise multi-view subspace clustering algorithm. However, classical graph learning methods suffer from high computational costs as they require maintaining the full $n \times n$ affinity or representation matrix, where n denotes the amount of data points. Besides, spectral clustering based on classical graph construction approaches will build a similarity graph of size $n \times n$ [8–11,25]. The computational complexity of this process is $\mathcal{O}(n^2)$, making the spectral clustering computationally expensive to construct graphs for the large-scale data. Obtaining eigenvectors from graphs through spectral embedding approaches [11,26] also demands considerable time and space costs. Overall, classical graph learning-based approaches encounter significant challenges in achieving satisfactory

* Corresponding author.
E-mail addresses: yc07426@um.edu.mo (C. Zhang), longchen@um.edu.mo (L. Chen), yc07456@um.edu.mo (Z. Shi), dwp9988@163.com (W. Ding).

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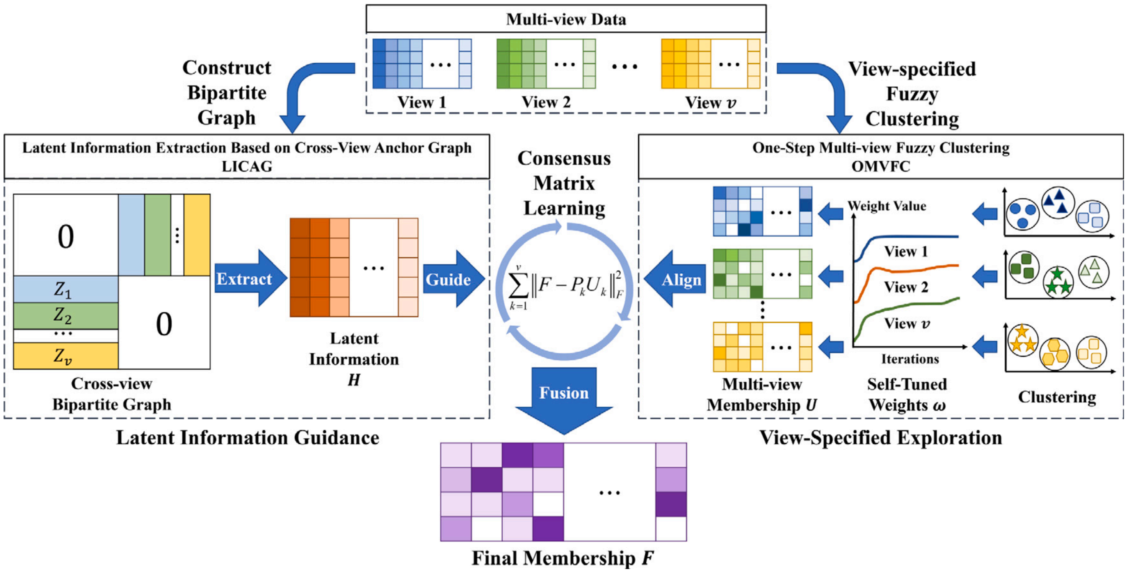

Fig. 1. The visualized depiction of the proposed multi-view fuzzy clustering method OMVFC-LICAG.

performance on large-scale datasets due to their high computational complexity in space and time.

To improve the computational efficiency of classical graph learning approaches, Nie et al. [27,28] put forward an anchor graph construction approach based on the bipartite graph learning algorithm. Recently, multi-view clustering frameworks have incorporated bipartite graph learning methods to reduce the computational cost when dealing with large-scale datasets [8,13,29–32]. Fang et al. [13] developed a unified framework (UDBGL) that simultaneously implemented view-consensus and view-specified bipartite graph learning. The UDBGL algorithm imposes the Laplacian rank constraint on the view-consensus matrix to directly obtain the cluster structure. Kang et al. [29] learned the self-expression matrix from anchor graphs of different views and achieved the linear time complexity (LMVSC). The paper [31] introduced a parameter-free subspace clustering algorithm, in which the self-expression matrix is learned from a consensus anchor graph (FPMVS-CAG). To efficiently obtain the low latent information of multi-view data, Wang et al. [32] performed singular value decomposition (SVD) on the sparse affinity matrix of anchors and data points. In this work, we also employ the bipartite graph learning technique to develop a novel cross-view anchor graph construction approach.

The K-means algorithm demonstrates outstanding performance and computational efficiency in a variety of applications. Researchers have worked on extending this algorithm to the field of multi-view clustering to exploit its advantages. Cai et al. [33] raised a robust multi-view K-means clustering algorithm that imposes the $\mathcal{L}_{2,1}$-norm on the cost function to achieve promising results. To improve the robustness of the multi-view K-means framework, Xing et al. [20] proposed a novel mixture correntropy metric, and Chen et al. [19] designed effective removal strategies to detect class-outliers and attribute-outliers. Additionally, graph learning approaches are also widely integrated into the multi-view K-means framework [5,18,34]. Nonetheless, the discrete constraint of the label indicator matrix in the K-means prototype poses optimization difficulties for the objective function, and the performance of the hard partition model may deteriorate when handling the complex data. To address these challenges, Liu et al. [35,36] proposed a late fusion mechanism to mix the partition matrix from different views, avoiding the optimization difficulties caused by the discrete constraint on the consensus label indicator matrix. Although the late fusion method slightly increases the complexity of the clustering procedure, it yields a more favorable performance.

Fuzzy theory is widely recognized as a powerful tool for modeling ambiguous and uncertain data. In multi-view data analysis, fuzzy clustering algorithms are also employed to overcome the aforementioned limitations of the multi-view K-means framework. Pedrycz [37] proposed the collaborative fuzzy clustering (CoFC) algorithm to handle multiple subsets of data patterns simultaneously. Cleuziou et al. introduced this technique into multi-view clustering and developed a centralized fuzzy clustering method, namely CoFKM [22]. It should be noted that the CoFKM algorithm treats all views equally, disregarding their intrinsic differences, which can lead to a decline in clustering performance. In response, Jiang et al. [38] utilized a view weighting approach to estimate the view-specified importance, while the work [21] further improved the collaborative fuzzy clustering framework with a feature weighting strategy. Wang et al. [39] integrated the minimax optimization with the fuzzy C-means algorithm and assigned larger weights to the corresponding views to suppress those with greater loss. To adaptively determine the diversity and importance of views, Zhu et al. [40] introduced a novel self-weighted approach to investigate and mix the distinct and common information across views. However, after learning the membership matrix of each view separately, most existing algorithms require additional post-processing steps to fuse the view-specified information and generate the final consensus matrix.

Despite the significant efforts made in the area of multi-view clustering, achieving a balance between view-specified membership exploration and consensus matrix learning during the optimization process remains a challenge. Furthermore, obtaining the final clustering result in a straightforward and effective manner is still an open question. Additionally, the potential benefits of combining multi-view fuzzy clustering with graph learning have not yet been fully investigated.

To address these challenges, we propose an efficient and effective multi-view fuzzy clustering algorithm, namely latent information-guided one-step multi-view fuzzy clustering based on cross-view anchor graph (OMVFC-LICAG). Specifically, the latent information is extracted from the cross-view anchor graph to preserve the global structure of the original data and improve the accuracy of multi-view clustering. Then, multi-view membership matrices are obtained from different views through the fuzzy clustering term, and the view-specified importance is determined by a novel self-tuned weighting mechanism. Moreover, the self-tuned weighting mechanism is used to balance view-specified membership exploration and consensus matrix learning during the optimization process. Lastly, the information of latent features and view-specified memberships is fused to generate the final consensus membership matrix without any post-processing step. The visualized depiction of our approach OMVFC-LICAG is illustrated in Fig. 1.

The main contributions of this work are summarized as follows.

Table 1
The main notations and their descriptions in this paper.

Notation	Description
$\{\mathbf{X}_k\}_{k=1}^{v}$	Multi-view data with v views
v	Number of views
$\mathbf{X}_k$	Feature matrix of the kth view, $\mathbf{X}_k \in \mathbb{R}^{q\times n}$
q	Number of feature dimensions
n	Number of data points
$\mathbf{x}_{j,k}$	The jth data point in the kth view
$\mathbf{C}_k$	Centroid matrix for the kth view, $\mathbf{C}_k \in \mathbb{R}^{q\times m}$
m	Number of clusters
$\mathbf{c}_{i,k}$	The ith centroid in the kth view
$\mathbf{U}_k$	Membership matrix of the kth view, $\mathbf{U}_k \in \mathbb{R}^{m\times n}$
$\mathbf{u}_{:,j,k}$	The jth column vector of the matrix $\mathbf{U}_k$
$\mathbf{P}_k$	Transition matrix for the kth view
$\mathbf{A}_k$	Anchors of the kth view
$rate$	Anchor rate of the cross-view anchor graph
$\mathbf{F}$	Consensus matrix, $\mathbf{F} \in \mathbb{R}^{m\times n}$
$\mathbf{H}$	Latent information feature from the cross-view anchor graph, $\mathbf{H} \in \mathbb{R}^{q_h\times n}$
q_h	The dimensionality of the latent information feature $\mathbf{H}$
$\mathbf{I}$	Identity matrix
$\mathbf{1}_m$	Column vector of size m with all elements 1
$\|\cdot\|_F$	Frobenius norm, $\|\mathbf{A}\|_F = \sqrt{\mathrm{Tr}\left(\mathbf{A}^\top\mathbf{A}\right)}$
$\mathrm{Tr}\left(\cdot\right)$	Trace of a square matrix
$[\cdot]_{m\times n}$	Matrix with size $m \times n$
$(\cdot)_{i,j,k}$	Element in the ith row and jth column of the kth matrix (view)

1. The proposed method generates the final clustering result through a straightforward one-step scheme that simultaneously explores view-specified membership matrices and learns the consensus matrix. Consequently, our approach eliminates the need for post-processing steps and effectively exploits both the raw and latent information of multi-view data.
2. To reduce the computational costs of graph-based approaches in terms of both time and space, we propose a novel approach for cross-view anchor graph construction based on bipartite graph learning. The low latent feature extracted from the graph is then employed to guide the optimization of the consensus matrix.
3. We introduce a self-tuned weighting mechanism to adaptively determine the importance of different views and to balance the interaction between view-specified membership exploration and consensus matrix learning. An ablation study demonstrates that our approach featuring the self-tuned weighting mechanism outperforms the fixed-weight method on all benchmark datasets.
4. The extensive experimental results provide compelling evidence that our approach surpasses the state-of-the-art approaches and achieves a favorable balance between clustering performance and efficiency. The code and datasets are available at https://github.com/ChuanbinZhang/OMVFC-LICAG.

The subsequent sections of this paper are organized as follows. In Section 2, we provide an introduction to the preliminaries and present related works on multi-view clustering and anchor-based graph learning algorithms. Section 3 elaborates on the proposed methods. Section 4 reports a comprehensive experimental analysis. Finally, Section 5 draws conclusions and discusses future works.

2. Preliminaries and related works

2.1. Notations

This paper adopts the following notation conventions: matrices are represented by boldface uppercase letters, such as $\mathbf{X}$; vectors are denoted by boldface lowercase letters, such as $\mathbf{x}$; and scalar values are expressed using lowercase letters, such as n. In addition, Table 1 presents some main notations and their corresponding descriptions used in this paper.

2.2. Multi-view clustering

This subsection will briefly review the previous literature on multi-view clustering, including spectral clustering-based, K-means-based and fuzzy clustering-based methods.

2.2.1. Multi-view spectral clustering

In the multi-view spectral clustering framework, researchers attempt to generate a consensus matrix $\mathbf{F}$ from different views via spectral clustering [8,25]. The objective function of the multi-view spectral clustering algorithm can be expressed as

$$\begin{aligned}&\min_{\mathbf{F},\mathbf{w}} \sum_{k=1}^{v} \left(w_k\right)^{\alpha} \mathrm{Tr}\left(\mathbf{F}\mathbf{L}_k\mathbf{F}^\top\right)\\ &\text{s.t.} \mathbf{F}\mathbf{F}^\top = \mathbf{I}, \mathbf{w}^\top\mathbf{1} = 1, \mathbf{w} \geq 0,\end{aligned} \tag{1}$$

where $\mathbf{L}_k$ denotes the normalized Laplacian matrix of the kth view, $\mathbf{w}$ represents a set of weight coefficients for different views and α is an exponential scalar. After obtaining the consensus matrix $\mathbf{F}$, an additional post-processing step is required, typically the K-means algorithm, to generate the final label assignment matrix. However, the space complexity of the classical graph learning approach is $\mathcal{O}(n^2)$ for a dataset of size n, making the generation of Laplacian matrices for large-scale datasets computationally expensive.

2.2.2. Multi-view K-means clustering

The extended versions of the K-means algorithm have been widely applied in multi-view clustering tasks [18,19,41]. Cai et al. [33] raised the robust multi-view K-means clustering (RMKMC) algorithm that solves the following problem:

$$\begin{aligned}\min \mathcal{J}_{\text{RMKMC}}\left(\mathbf{F},\mathbf{w}\right) = \sum_{k=1}^{v} \left(w_k\right)^{\alpha} &\left\|\mathbf{X}_k^\top - \mathbf{F}^\top\mathbf{C}_k^\top\right\|_{2,1}\\ \text{s.t.} \mathbf{F}^\top\mathbf{1} &= \mathbf{1}, \mathbf{F} \in \{0,1\},\\ \mathbf{w}^\top\mathbf{1} &= 1, \mathbf{w} > 0,\end{aligned} \tag{2}$$

where $\mathbf{F} \in \mathbb{R}^{m\times n}$ is an indicator matrix representing the cluster assignment, $\mathbf{C}_k \in \mathbb{R}^{q\times m}$ denotes the centroid matrix of the k-the view, and $\mathbf{w}$ is the set of weight coefficients. Unlike the RMKMC algorithm, which directly obtains the label assignment matrix, Liu et al. [35,36] proposed a late fusion mechanism to generate the consensus matrix by fusing the base partition results from different views. Suppose that a set of base clustering results, denoted by $\{\mathbf{B}_k\}_{k=1}^{v}$ with $\mathbf{B}_k \in \mathbb{R}^{m\times n}$, has been obtained for v views, the objective of the late fusion-based multi-view clustering (LFMVC) algorithm is expressed as

$$\begin{aligned}\max \mathcal{J}_{\text{LFMVC}}\left(\mathbf{F},\mathbf{C},\mathbf{P},\mathbf{w}\right) &= \mathrm{Tr}\left(\mathbf{C}^\top\mathbf{F}\sum_{k=1}^{v} w_k\mathbf{B}_k^\top\mathbf{P}_k\right)\\ \text{s.t.} \mathbf{C}^\top\mathbf{C} &= \mathbf{I}_m, \mathbf{P}_k^\top\mathbf{P}_k = \mathbf{I}_m, \forall k,\\ \mathbf{F} &\in \{0,1\}^{m\times n}, \sum_{k=1}^{v} w_k^2 = 1, \mathbf{w} \geq 0,\end{aligned} \tag{3}$$

where $\mathbf{C} \in \mathbb{R}^{m\times m}$ denotes the consensus centroid matrix and $\mathbf{P}_k$ represents the transition matrix for the corresponding kth view. Although multi-view K-means clustering is an efficient and effective framework, the discrete nature of the indicator matrix $\mathbf{F}$ makes it difficult to optimize, and the final clustering results may be crisp on complex datasets.

2.2.3. Multi-view fuzzy clustering

Fuzzy clustering refers to a family of methods for assigning data points to multiple clusters with membership degrees [42]. The label assignment matrix in the fuzzy clustering framework contains continuous variables that represent membership degree values ranging from 0 to 1. It is clear that the membership variable is more capable of expressing ambiguous relationships in the data compared to the discrete variable.

Cleuziou et al. [22] introduced the FCM algorithm into the multi-view clustering community and proposed a fuzzy clustering approach for multi-view data, namely CoFKM. The formulation of the CoFKM's objective function is

$$\min \mathcal{J}_{\text{CoFKM}}(\mathbf{U}, \mathbf{C}) = \sum_{k=1}^{v} \sum_{j=1}^{n} \sum_{i=1}^{m} \tilde{u}_{i,j,k} \left\| \mathbf{x}_{j,k} - \mathbf{c}_{i,k} \right\|_2^2, \tag{4}$$

where $\tilde{u}_{i,j,k} = (1-\alpha)u_{i,j,k}^{\gamma} + \frac{\alpha}{v-1}\sum_{l=1}^{l\neq k} u_{i,j,l}^{\gamma}$ is an auxiliary variable that allows the information to spread across views, $u_{i,j,k}$ denotes the membership degree and γ represents the fuzzification coefficient. The final membership values shared by all views are calculated by $\hat{u}_{i,j,k} = \sqrt[v]{\prod_{k=1}^{v} u_{i,j,k}}$. Subsequently, various machine learning approaches were integrated into the multi-view fuzzy clustering framework. For instance, novel weighting mechanisms were proposed to estimate the importance of views in some studies [22,40]. In addition to exploring the discriminability of views, some researchers imposed regularization terms on membership matrices to further enhance the effectiveness of multi-view fuzzy clustering approaches. Han et al. [23] proposed a weight allocation scheme to learn view-specified information of multi-view data and imposed the adaptive sparse constraint to the consensus membership matrix (MVASM). Besides, Wei et al. [43] imposed the low-rank regularization to the concatenated membership tensor and achieved a robust model for multi-view data. Nevertheless, the majority of current multi-view fuzzy clustering frameworks still require additional post-processing steps to generate a consensus membership matrix and cannot directly obtain the final clustering result.

2.3. Bipartite graph learning

Classical graph construction approaches are computationally expensive, as they require the maintenance of a full $n \times n$ matrix to represent pairwise relationships between samples. To reduce the computational complexity associated with building graphs for large-scale datasets, Nie et al. [27] proposed the bipartite graph learning approach, which is capable of representing indirect relationships of anchors and data points efficiently. Let $\mathbf{X} = \{\mathbf{x}_j\}_{j=1}^{n}$ be a single-view dataset comprising n data points, and $\mathbf{A} = \{\mathbf{a}_i\}_{i=1}^{r}$ denotes a set of anchors that can be generated by random sampling or clustering algorithms (e.g., K-means), where r is the number of anchors. The objective of bipartite graph learning is to minimize the following problem:

$$\begin{aligned} &\min_{\mathbf{Z}} \sum_{j=1}^{n} \sum_{i=1}^{r} \left(d(\mathbf{a}_i, \mathbf{x}_j) z_{i,j} + \lambda z_{i,j}^2 \right) \\ &\text{s.t.} \sum_{i=1}^{r} z_{i,j} = 1, z_{i,j} \geq 0, \forall 1 \leq j \leq n, \end{aligned} \tag{5}$$

where $d(\mathbf{a}_i, \mathbf{x}_j)$ represents the dissimilarity of the jth data point and the ith anchor, such as square Euclidean distance. $\mathbf{Z} \in \mathbb{R}^{r \times n}$ denotes the correlation matrix of the anchors $\mathbf{A}$ and the dataset $\mathbf{X}$. The solution of problem (5) is given by:

$$z_{i,j} = \begin{cases} \frac{d(\mathbf{x}_j, \mathbf{a}_{k+1}) - d(\mathbf{x}_j, \mathbf{a}_i)}{k d(\mathbf{x}_j, \mathbf{a}_{k+1}) - \sum_{l=1}^{k} d(\mathbf{x}_j, \mathbf{a}_l)}, & i \leq k \\ 0, & i > k \end{cases} \tag{6}$$

where k denotes the k-nearest anchors of the jth data point. Finally, the bipartite graph is represented by a similarity matrix $\mathbf{W} \in \mathbb{R}^{(n+r)\times(n+r)}$, which is defined as follows:

$$\mathbf{W} = \begin{bmatrix} \mathbf{0} & \mathbf{Z}^{\top} \\ \mathbf{Z} & \mathbf{0} \end{bmatrix}. \tag{7}$$

To take advantages of bipartite graph learning, we develop a cross-view anchor graph approach that boosts the efficiency and effectiveness of the multi-view fuzzy clustering framework.

3. Methodology

In this paper, we develop a novel multi-view fuzzy clustering algorithm that possesses the following properties.

1. **Simplicity**: The proposed method allows for direct acquisition of final clustering results. Additionally, the fusion of the view-specified information is implemented in a computationally efficient manner.
2. **Self-tuning**: In our approach, the view-specified membership exploration and consensus matrix learning operations are performed simultaneously, and a novel self-tuned weighting mechanism is employed to adaptively balance these two operations during the optimization process.
3. **Scalability**: By employing bipartite graph learning, we can generate a cross-view anchor graph for multi-view data, making the graph construction approach more efficient and feasible in terms of both time and space, particularly for large-scale datasets.

The proposed method, as depicted in Fig. 1, is composed of three main components: latent information guidance, view-specified exploration and consensus matrix learning. First, the latent information guidance extracts the low-dimensional feature in the cross-view anchor graph and guides the optimization of the consensus matrix from a global perspective. Second, view-specified exploration optimizes single-view membership matrices and autonomously determines weights of diverse views. Third, consensus matrix learning fuses all the above features and directly generates the final clustering result in an efficient and concise manner.

3.1. One-step multi-view fuzzy clustering (OMVFC)

To achieve the simplicity and self-tuning properties of our framework, we design a one-step scheme for multi-view fuzzy clustering, namely OMVFC. In the OMVFC algorithm, the consensus membership matrix is obtained through the linear transformation and combination of view-specified membership matrices, which eliminates the need for an additional post-processing step. The self-tuned weighting mechanism investigates the membership quality of each view at a fine-grained level and adaptively determines the impact of various views on the consensus matrix. This allows our approach to fully explore single-view data and automatically optimize the fusion of different features.

3.1.1. Problem formulation of OMVFC

The OMVFC algorithm is implemented by minimizing the following problem:

$$\begin{aligned} &\min_{\mathbf{U}_k, \mathbf{C}_k, \mathbf{P}_k, \mathbf{F}} \sum_{k=1}^{v} \sqrt{\sum_{j=1}^{n} \sum_{i=1}^{m} \frac{1}{q_k} \left\| \mathbf{x}_{j,k} - \mathbf{c}_{i,k} \right\|_2^2 u_{i,j,k} + \frac{1}{\alpha} \left\| \mathbf{F} - \mathbf{P}_k \mathbf{U}_k \right\|_F^2} \\ \Leftrightarrow &\min_{\mathbf{U}_k, \mathbf{C}_k, \mathbf{P}_k, \mathbf{F}} \sum_{k=1}^{v} \sqrt{\alpha \operatorname{Tr}\left(\mathbf{D}_k^{\top} \mathbf{U}_k\right) + \left\| \mathbf{F} - \mathbf{P}_k \mathbf{U}_k \right\|_F^2} \\ &\quad \text{s.t. } \mathbf{F}^{\top} \mathbf{1}_m = \mathbf{1}_n, \mathbf{F} \geq 0, \mathbf{P}_k^{\top} \mathbf{P}_k = \mathbf{I}, \\ &\quad\quad \mathbf{U}_k^{\top} \mathbf{1}_m = \mathbf{1}_n, \mathbf{U}_k \geq 0, \forall 1 \leq k \leq v, \end{aligned} \tag{8}$$

where $\mathbf{D}_k = \left[d_{i,j,k}\right]_{c \times n}, d_{i,j,k} = \frac{1}{q_k} \left\| \mathbf{x}_{j,k} - \mathbf{c}_{i,k} \right\|_2^2$ calculates the dissimilarity of a data point and the centroid, q_k is the feature dimensions of the kth view and α denotes a trade-off hyperparameter. It is noteworthy that the factor $1/q_k$ effectively mitigates disparities in the range of dissimilarity values due to the number of dimensions, thereby facilitating the fine-tuning of parameters and enhancing the generalization of parameter selection. Inspired by [44,45], we introduce self-tuned weights $\{\omega_k\}_{k=1}^{v}$ for each view into the objective function (8), which estimates the importance of different views adaptively. Finally, the problem (8) is reformulated as

$$\begin{aligned}
&\min_{\mathbf{U}_k,\mathbf{C}_k,\mathbf{P}_k,\mathbf{F}} \underbrace{\sum_{k=1}^{v}\omega_k}_{\substack{\text{Self-Tuned}\\ \text{Weighting}}} \left(\alpha \underbrace{\operatorname{Tr}\left(\mathbf{D}_k^\top \mathbf{U}_k\right)}_{\substack{\text{View-Specified}\\ \text{Exploration}}} + \underbrace{\left\|\mathbf{F}-\mathbf{P}_k\mathbf{U}_k\right\|_F^2}_{\substack{\text{Consensus Matrix}\\ \text{Learning}}} \right)\\
\Leftrightarrow &\min_{\mathbf{U}_k,\mathbf{C}_k,\mathbf{P}_k,\mathbf{F}} \sum_{k=1}^{v}\omega_k \left[\alpha \operatorname{Tr}\left(\mathbf{D}_k^\top \mathbf{U}_k\right) + \operatorname{Tr}\left(\mathbf{F}_k^\top\mathbf{F}_k - 2\mathbf{F}^\top\mathbf{P}_k\mathbf{U}_k + \mathbf{U}_k^\top\mathbf{U}_k\right)\right]\\
\Leftrightarrow &\min_{\mathbf{U}_k,\mathbf{C}_k,\mathbf{P}_k,\mathbf{F}} \sum_{k=1}^{v}\omega_k \operatorname{Tr}\left(\alpha\mathbf{D}_k^\top \mathbf{U}_k + \mathbf{F}_k^\top\mathbf{F}_k - 2\mathbf{F}^\top\mathbf{P}_k\mathbf{U}_k + \mathbf{U}_k^\top\mathbf{U}_k\right)\\
&\quad \text{s.t. } \mathbf{F}^\top\mathbf{1}_m = \mathbf{1}_n, \mathbf{F}\geq 0, \mathbf{P}_k^\top\mathbf{P}_k = \mathbf{I}_m,\\
&\quad\quad \mathbf{U}_k^\top\mathbf{1}_m = \mathbf{1}_n, \mathbf{U}_k \geq 0, \forall 1\leq k\leq v.
\end{aligned} \tag{9}$$

The proposed OMVFC algorithm contains two terms in its objective function (9): the view-specified exploration term, which learns the intrinsic cluster structure of each view, and the consensus matrix learning term, which fuses membership matrices from different views. Additionally, when the objective function (9) is minimized, the self-tuned weight ω_k is

$$\omega_k = \frac{1}{2\sqrt{\operatorname{Tr}\left(\alpha\mathbf{D}_k^\top\mathbf{U}_k + \mathbf{F}_k^\top\mathbf{F}_k - 2\mathbf{F}^\top\mathbf{P}_k\mathbf{U}_k + \mathbf{U}_k^\top\mathbf{U}_k\right)}}. \tag{10}$$

At the early stage of optimization, the self-tuned weighting mechanism prevents the poor quality information of view-specified membership matrices from spreading among the views. This helps to avoid performance degradation and unstable clustering results.

3.1.2. Optimization of OMVFC

In this subsection, we present an effective update scheme to solve the problem (9) using the alternating optimization algorithm.

C-update: by fixing $\{\mathbf{U}_k\}_{k=1}^v$, $\mathbf{F}$ and $\{\mathbf{P}_k\}_{k=1}^v$, the minimization of Eq. (9) with respect to $\mathbf{c}_{i,k}$ can be rewritten as

$$\min_{\mathbf{c}_{i,k}} \sum_{j=1}^{n}\sum_{i=1}^{m} \left\|\mathbf{x}_{j,k} - \mathbf{c}_{i,k}\right\|_2^2 u_{i,j,k}. \tag{11}$$

When we take the derivative of Eq. (11) with respect to $\mathbf{c}_{i,k}$ and equate it to zero, the following equation holds:

$$\sum_{j=1}^{n} u_{i,j,k}\left(\mathbf{x}_{j,k} - \mathbf{c}_{i,k}\right) = 0. \tag{12}$$

Then, the $\mathbf{c}_{i,k}$ is updated by

$$\mathbf{c}_{i,k} = \frac{\sum_{j=1}^{n} u_{i,j,k}\mathbf{x}_{j,k}}{\sum_{j=1}^{n} u_{i,j,k}}. \tag{13}$$

P-update: by fixing $\{\mathbf{U}_k\}_{k=1}^v$, $\{\mathbf{C}_k\}_{k=1}^v$ and $\mathbf{F}$, the minimization of Eq. (9) with respect to $\mathbf{P}_k$ is transformed into

$$\begin{aligned}
&\min_{\mathbf{P}_k} \operatorname{Tr}\left(-2\mathbf{U}_k\mathbf{F}^\top\mathbf{P}_k\right)\\
\Leftrightarrow &\max_{\mathbf{P}_k} \operatorname{Tr}\left(\mathbf{U}_k\mathbf{F}^\top\mathbf{P}_k\right)\\
&\text{s.t. } \mathbf{P}_k^\top\mathbf{P}_k = \mathbf{I}_m, \forall 1\leq k\leq v.
\end{aligned} \tag{14}$$

Let $\mathbf{Q}_k = \mathbf{U}_k\mathbf{F}^\top$, and problem (14) can be solved by using a SVD-based method in [46]. By decomposing $\mathbf{Q}_k = \mathbf{S}_k\mathbf{\Sigma}_k\mathbf{V}_k^\top$, the solution of $\mathbf{P}_k$ is

$$\mathbf{P}_k = \mathbf{V}_k\mathbf{S}_k^\top. \tag{15}$$

U-update: by fixing $\{\mathbf{C}_k\}_{k=1}^v$, $\mathbf{F}$ and $\{\mathbf{P}_k\}_{k=1}^v$, the minimization of Eq. (9) with respect to $\mathbf{U}_k$ is reformulated as

$$\begin{aligned}
&\min_{\mathbf{U}_k} \operatorname{Tr}\left(\alpha\mathbf{D}_k^\top\mathbf{U}_k - 2\mathbf{F}^\top\mathbf{P}_k\mathbf{U}_k + \mathbf{U}_k^\top\mathbf{U}_k\right)\\
\Leftrightarrow &\min_{\mathbf{U}_k} \operatorname{Tr}\left(\mathbf{U}_k^\top\mathbf{U}_k - 2\frac{2\mathbf{F}^\top\mathbf{P}_k - \alpha\mathbf{D}_k^\top}{2}\mathbf{U}_k\right)\\
&\text{s.t. } \mathbf{U}_k^\top\mathbf{1}_m = \mathbf{1}_n, \mathbf{U}_k\geq 0, \forall 1\leq k\leq v.
\end{aligned} \tag{16}$$

By denoting $\mathbf{E}_k^\top = \left(2\mathbf{F}^\top\mathbf{P}_k - \alpha\mathbf{D}_k^\top\right)/2 = \left[e_{j,i,k}\right]_{n\times m}$, the problem (16) can be expressed as

$$\begin{aligned}
&\min_{\mathbf{U}_k} \operatorname{Tr}\left(\mathbf{U}_k^\top\mathbf{U}_k - 2\mathbf{E}_k^\top\mathbf{U}_k\right)\\
\Leftrightarrow &\min_{\mathbf{U}_k} \left\|\mathbf{U}_k - \mathbf{E}_k\right\|_F^2\\
\Leftrightarrow &\min_{\mathbf{u}_{:,j,k}} \sum_{j=1}^{n}\left\|\mathbf{u}_{:,j,k} - \mathbf{e}_{:,j,k}\right\|_2^2\\
&\text{s.t. } \mathbf{u}_{:,j,k}^\top\mathbf{1}_m = 1, \mathbf{u}_{:,j,k}\geq 0, \forall 1\leq j\leq n,
\end{aligned} \tag{17}$$

where $\mathbf{u}_{:,j,k}$ and $\mathbf{e}_{:,j,k}$ denote the jth column vectors of $\mathbf{U}_k$ and $\mathbf{E}_k$, respectively. It is clear that the $\mathbf{u}_{:,j,k}$ is independent for each j, the problem (17) can be decomposed into the following sub-problems column-wise [23]:

$$\begin{aligned}
&\min_{\mathbf{u}_{:,j,k}} \frac{1}{2}\left\|\mathbf{u}_{:,j,k} - \mathbf{e}_{:,j,k}\right\|_2^2\\
&\text{s.t. } \mathbf{u}_{:,j,k}^\top\mathbf{1}_m = 1, \mathbf{u}_{:,j,k}\geq 0,
\end{aligned} \tag{18}$$

The Lagrangian function of Eq. (18) is

$$\mathcal{L}\left(\mathbf{u}_{:,j,k},\gamma_j,\boldsymbol{\eta}_j\right) = \frac{1}{2}\left\|\mathbf{u}_{:,j,k} - \mathbf{e}_{:,j,k}\right\|_2^2 - \gamma_j\left(\mathbf{u}_{:,j,k}^\top\mathbf{1} - 1\right) - \boldsymbol{\eta}_j^\top\mathbf{u}_{:,j,k}, \tag{19}$$

where the scalar γ_j and the vector $\boldsymbol{\eta}_j$ are Lagrangian multipliers corresponding to constraints in problem (18). The Karush–Kuhn–Tucker (KKT) conditions are formulated as below.

$$\begin{cases}
\forall 1\leq i\leq m, & u_{i,j,k} - e_{i,j,k} - \gamma_j - \eta_{i,j} = 0\\
\forall 1\leq i\leq m, & \eta_{i,j}u_{i,j,k} = 0\\
\forall 1\leq i\leq m, & u_{i,j,k}\geq 0, \eta_{i,j}\geq 0
\end{cases} \tag{20}$$

By combining with the constraint $\mathbf{u}_{:,j,k}^\top\mathbf{1} = 1$, we have

$$\mathbf{u}_{:,j,k} = \max\left(\mathbf{e}_{:,j,k} - \frac{\mathbf{1}\mathbf{1}^\top}{m}\mathbf{e}_{:,j,k} + \frac{1}{m}\mathbf{1} - \bar{\eta}_j\mathbf{1}, 0\right), \tag{21}$$

where $\bar{\eta}_j$ denotes the average of $\boldsymbol{\eta}_j$ and is obtained by using the algorithm in [23].

F-update: by fixing $\{\mathbf{U}_k\}_{k=1}^v$, $\{\mathbf{C}_k\}_{k=1}^v$ and $\{\mathbf{P}_k\}_{k=1}^v$, the minimization of Eq. (9) with respect to $\mathbf{F}$ is simplified to

$$\begin{aligned}
&\min_{\mathbf{F}} \sum_{k=1}^{v}\omega_k \operatorname{Tr}\left(\mathbf{F}^\top\mathbf{F} - 2\mathbf{F}^\top\mathbf{P}_k\mathbf{U}_k\right)\\
\Leftrightarrow &\min_{\mathbf{F}} \operatorname{Tr}\left(\sum_{k=1}^{v}\omega_k\mathbf{F}^\top\mathbf{F} - 2\sum_{k=1}^{v}\omega_k\mathbf{U}_k^\top\mathbf{P}_k^\top\mathbf{F}\right)\\
\Leftrightarrow &\min_{\mathbf{F}} \operatorname{Tr}\left(\mathbf{F}^\top\mathbf{F} - 2\frac{\sum_{k=1}^{v}\omega_k\mathbf{U}_k^\top\mathbf{P}_k^\top}{\sum_{k=1}^{v}\omega_k}\mathbf{F}\right)\\
\Leftrightarrow &\min_{\mathbf{F}} \left\|\mathbf{F} - \mathbf{G}\right\|_2^2\\
\Leftrightarrow &\min_{\mathbf{f}_{:,j}} \sum_{j=1}^{n}\left\|\mathbf{f}_{:,j} - \mathbf{g}_{:,j}\right\|_2^2\\
&\text{s.t. } \mathbf{F}^\top\mathbf{1}_m = \mathbf{1}_n, \mathbf{F}\geq 0,
\end{aligned} \tag{22}$$

where

$$\mathbf{G} = \left[g_{i,j}\right]_{m\times n}, g_{i,j} = \frac{\sum_{k=1}^{v}\omega_k\mathbf{U}_k^\top\mathbf{P}_k^\top}{\sum_{k=1}^{v}\omega_k}. \tag{23}$$

In a manner similar to the updating step of $\mathbf{U}_k$, we solve the problem (22) by minimizing the sub-problem as follows:

$$\begin{aligned}
&\min_{\mathbf{f}_{:,j}} \frac{1}{2}\left\|\mathbf{f}_{:,j} - \mathbf{g}_{:,j}\right\|_2^2\\
&\text{s.t. } \mathbf{f}_{:,j}^\top\mathbf{1}_m = 1, \mathbf{f}_{:,j}\geq 0,
\end{aligned} \tag{24}$$

where $\mathbf{f}_{:,j}$ and $\mathbf{g}_{:,j}$ are the jth column vectors of $\mathbf{F}$ and $\mathbf{G}$, respectively. In a nutshell, the $\mathbf{f}_{:,j}$ is updated by

$$\mathbf{f}_{:,j} = \max\left(\mathbf{g}_{:,j} - \frac{\mathbf{1}\mathbf{1}^\top}{m}\mathbf{g}_{:,j} + \frac{1}{m}\mathbf{1} - \bar{\eta}_j\mathbf{1}, 0\right), \tag{25}$$

where the definition of $\bar{\eta}_j$ is similar to that in Eq. (21).

3.2. Latent information-guided one-step multi-view fuzzy clustering based on cross-view anchor graph (OMVFC-LICAG)

3.2.1. Latent information extraction based on cross-view anchor graph (LICAG)

To achieve the scalability in our method and boost the performance of the OMVFC algorithm, we introduce the latent features extracted from a cross-view anchor graph into our framework. Specifically, we construct similarity matrices for each view $\{\mathbf{Z}_k\}_{k=1}^{v}$ by performing bipartite graph learning on the single-view data, as elaborated in Section 2.3. Here, $\mathbf{Z}_k \in \mathbb{R}^{r\times n}$ denotes the similarity matrix of data points and anchors in the kth view, where r represents the number of anchors. The amount of anchors is determined by the anchor rate such that $r = n \times rate$, and we use the K-means algorithm to generate the anchor points. Finally, the similarity matrix of the cross-view anchor graph $\mathbf{W} \in \mathbb{R}^{(n+v\cdot r)\times(n+v\cdot r)}$ is represented as follows:

$$\mathbf{W} = \begin{bmatrix} \mathbf{0} & \mathbf{Z}_1^\top & \cdots & \mathbf{Z}_v^\top \\ \mathbf{Z}_1 & & & \\ \vdots & & \mathbf{0} & \\ \mathbf{Z}_v & & & \end{bmatrix}, \tag{26}$$

where v is the amount of views. Then, the normalized Laplacian matrix of the cross-view anchor graph $\mathbf{W}$ is calculated by $\mathbf{L} = \mathbf{I} - \mathbf{\Delta}^{-\frac{1}{2}}\mathbf{W}\mathbf{\Delta}^{-\frac{1}{2}}$, where $\mathbf{\Delta}$ represents a diagonal matrix such that $\mathbf{\Delta}_{i,i} = \sum_j \mathbf{w}_{i,j}$. Next, the spectral embedding of the cross-view anchor graph, denoted by $\widetilde{\mathbf{H}}$, can be obtained by solving the following problem:

$$\begin{aligned} &\min_{\widetilde{\mathbf{H}}} \mathrm{Tr}\left(\widetilde{\mathbf{H}}\mathbf{L}\widetilde{\mathbf{H}}^\top\right) \\ &\text{s.t. } \widetilde{\mathbf{H}}\widetilde{\mathbf{H}}^\top = \mathbf{I}. \end{aligned} \tag{27}$$

The solution for $\widetilde{\mathbf{H}}$ is a matrix consists of eigenvectors that correspond to the q_h smallest eigenvalues, where q_h is the user-specified dimension. We consider $\widetilde{\mathbf{H}}$ as a concatenated spectral embedding matrix, which can be denoted as $\widetilde{\mathbf{H}} = \left[\widetilde{\mathbf{H}}_{\mathbf{X}} \,\middle|\, \widetilde{\mathbf{H}}_{\mathbf{A}}\right]$. Here, $\widetilde{\mathbf{H}}_{\mathbf{X}}$ and $\widetilde{\mathbf{H}}_{\mathbf{A}}$ represent spectral embedding matrices of raw data points and anchors, respectively. $\mathbf{H} = \widetilde{\mathbf{H}}_{\mathbf{X}} \in \mathbb{R}^{q_h\times n}$ is a low feature matrix extracted from the cross-view anchor graph, and can be considered as the fused latent information of multi-view data. To generate the label assignment matrix, we can perform the K-means algorithm on $\mathbf{H}$, which is referred to as the LICAG algorithm. Moreover, we integrate the extracted latent feature $\mathbf{H}$ with our approach OMVFC to achieve better clustering performance, resulting in the ultimate method called OMVFC-LICAG.

3.2.2. Problem formulation of OMVFC-LICAG

In the end, we append a latent information guidance term to Eq. (9) and formulate the final objective of the OMVFC-LICAG algorithm as follows:

$$\begin{aligned} &\min_{\mathbf{U}_k,\mathbf{C}_k,\mathbf{P}_k,\mathbf{F},\mathbf{O}} \underbrace{\sum_{k=1}^{v}\omega_k}_{\substack{\text{Self-Tuned}\\ \text{Weighting}}} \left(\alpha \underbrace{\mathrm{Tr}\left(\mathbf{D}_k^\top \mathbf{U}_k\right)}_{\substack{\text{View-Specified}\\ \text{Exploration}}} + \underbrace{\left\|\mathbf{F} - \mathbf{P}_k\mathbf{U}_k\right\|_F^2}_{\substack{\text{Consensus Matrix}\\ \text{Learning}}} \right) \\ &\qquad + \beta \underbrace{\sum_{j=1}^{n}\sum_{i=1}^{m} \left\|\mathbf{h}_j - \mathbf{o}_i\right\|_2^2 f_{i,j}}_{\text{Latent Information Guidance}} \\ \Leftrightarrow &\min_{\mathbf{U}_k,\mathbf{C}_k,\mathbf{P}_k,\mathbf{F},\mathbf{O}} \sum_{k=1}^{v}\omega_k \,\mathrm{Tr}\left(\alpha\mathbf{D}_k^\top\mathbf{U}_k + \mathbf{F}_k^\top\mathbf{F}_k - 2\mathbf{F}^\top\mathbf{P}_k\mathbf{U}_k + \mathbf{U}_k^\top\mathbf{U}_k\right) \\ &\qquad + \beta\,\mathrm{Tr}\left(\mathbf{D}_H^\top\mathbf{F}\right) \\ &\qquad \text{s.t. } \mathbf{F}^\top\mathbf{1}_m = \mathbf{1}_n, \mathbf{F} \geq 0, \mathbf{P}_k^\top\mathbf{P}_k = \mathbf{I}_m, \\ &\qquad\qquad \mathbf{U}_k^\top\mathbf{1}_m = \mathbf{1}_n, \mathbf{U}_k \geq 0, \forall 1 \leq k \leq v, \end{aligned} \tag{28}$$

where $\mathbf{D}_H = \left[d_{i,j,h}\right]_{c\times n}, d_{i,j,h} = \left\|\mathbf{h}_j - \mathbf{o}_i\right\|_2^2$, and ω_k is given in Eq. (10). $\mathbf{H} \in \mathbb{R}^{q_h\times n}$ represents the latent features of the multi-view data, $\mathbf{o}_i$ denotes the ith centroid in latent space and q_h is the dimensionality of latent features. The latent information guidance term in the objective function (28) serves to provide the cross-view knowledge for our framework, which enables the search for a relatively good solution to the consensus matrix from a global perspective.

3.2.3. Optimization of OMVFC-LICAG

As the solutions of $\{\mathbf{U}_k\}_{k=1}^{v}$, $\{\mathbf{C}_k\}_{k=1}^{v}$, and $\{\mathbf{P}_k\}_{k=1}^{v}$ in the objective function (28) are independent of the latent feature matrix $\mathbf{H}$, they can be solved in the same way as the schemes in problem (9). To avoid being verbose, we present only the updating steps for the centroid matrix $\mathbf{O}$ of latent futures and the consensus membership $\mathbf{F}$.

O-update: by fixing $\{\mathbf{U}_k\}_{k=1}^{v}$, $\{\mathbf{C}_k\}_{k=1}^{v}$, $\{\mathbf{P}_k\}_{k=1}^{v}$ and $\mathbf{F}$ the minimization of Eq. (28) with respect to $\mathbf{o}_i$ is rewritten as

$$\min_{\mathbf{o}_i} \sum_{j=1}^{n}\sum_{i=1}^{m} \left\|\mathbf{h}_j - \mathbf{o}_i\right\|_2^2 f_{i,j}. \tag{29}$$

When we take the derivative of Eq. (29) with respect to $\mathbf{o}_i$ and equate it to zero, the following equation holds:

$$\sum_{j=1}^{n} f_{i,j}\left(\mathbf{h}_j - \mathbf{o}_i\right) = 0. \tag{30}$$

Then, the $\mathbf{o}_i$ is updated by

$$\mathbf{o}_i = \frac{\sum_{j=1}^{n} f_{i,j}\mathbf{h}_j}{\sum_{j=1}^{n} f_{i,j}}. \tag{31}$$

F-update: by fixing variables other than $\mathbf{F}$, the problem (28) is simplified to

$$\begin{aligned} &\min_{\mathbf{F}} \mathrm{Tr}\left(\sum_{k=1}^{v}\omega_k\mathbf{F}^\top\mathbf{F} - 2\sum_{k=1}^{v}\omega_k\mathbf{U}_k^\top\mathbf{P}_k^\top\mathbf{F} + \beta\mathbf{D}_H^\top\mathbf{F}\right) \\ \Leftrightarrow &\min_{\mathbf{F}} \mathrm{Tr}\left(\sum_{k=1}^{v}\omega_k\mathbf{F}^\top\mathbf{F} - 2\left(\sum_{k=1}^{v}\omega_k\mathbf{U}_k^\top\mathbf{P}_k^\top - \frac{\beta}{2}\mathbf{D}_H^\top\right)\mathbf{F}\right) \\ \Leftrightarrow &\min_{\mathbf{F}} \mathrm{Tr}\left(\mathbf{F}^\top\mathbf{F} - 2\mathbf{G}^\top\mathbf{F}\right) \\ \Leftrightarrow &\min_{\mathbf{F}} \|\mathbf{F} - \mathbf{G}\|_F^2 \\ \Leftrightarrow &\min_{\mathbf{f}_{:,j}} \sum_{j=1}^{n} \left\|\mathbf{f}_{:,j} - \mathbf{g}_{:,j}\right\|_2^2 \\ &\text{s.t. } \mathbf{F}^\top\mathbf{1}_m = \mathbf{1}_n, \mathbf{F} \geq 0, \end{aligned} \tag{32}$$

where

$$\mathbf{G}^\top = \frac{\sum_{k=1}^{v}\omega_k\mathbf{U}_k^\top\mathbf{P}_k^\top - \frac{\beta}{2}\mathbf{D}_H^\top}{\sum_{k=1}^{v}\omega_k}. \tag{33}$$

Since the $\mathbf{f}_{:,j}$ is independent for each j, we can solve the sub-problem for each $\mathbf{f}_{:,j}$ to address problem (32):

$$\begin{aligned} &\min_{\mathbf{f}_{:,j}} \frac{1}{2}\left\|\mathbf{f}_{:,j} - \mathbf{g}_{:,j}\right\|_2^2 \\ &\text{s.t. } \mathbf{f}_{:,j}^\top\mathbf{1}_m = 1, \mathbf{f}_{:,j} \geq 0, \end{aligned} \tag{34}$$

Finally, the $\mathbf{f}_{:,j}$ is updated by

$$\mathbf{f}_{:,j} = \max\left(\mathbf{g}_{:,j} - \frac{\mathbf{1}\mathbf{1}^\top}{m}\mathbf{g}_{:,j} + \frac{1}{m}\mathbf{1} - \bar{\eta}_j\mathbf{1}, 0\right), \tag{35}$$

where the definition of $\bar{\eta}_j$ is similar to that in Eq. (21).

To sum up, the pseudo-code of the OMVFC-LICAG algorithm is described in Algorithm 1.

3.3. Computational complexity analysis

The proposed method comprises two phases. The first phase involves a pre-processing operation that extracts the latent information from the cross-view anchor graph. The second phase involves a clustering process that integrates the proposed one-step multi-view

Algorithm 1 OMVFC-LICAG

Input: Multi-view data $\{\mathbf{X}_k\}_{k=1}^{v}$, the trade-off hyperparameters α and β, the stopping threshold value ϵ, and the maximum iterations t_{max}.
Output: The final clustering result.
1: Generate the latent feature matrix $\mathbf{H}$ via the LICAG algorithm;
2: Set the transition matrices $\{\mathbf{P}_k^{(0)}\}_{k=1}^{v} = \mathbf{I}$ and $t = 0$;
3: Initialize the consensus matrix $\mathbf{F}^{(0)}$ and view-specified membership $\mathbf{U}_k^{(0)}$ by performing the classical fuzzy C-means algorithm on the latent feature matrix $\mathbf{H}$ and each view's data matrix $\mathbf{X}_k$, respectively;
4: Calculate the self-tuned weights $\{\omega_k^{(0)}\}_{k=1}^{v}$ by using Eq. (10);
5: **repeat**
6: **for** $k = 1 : v$ **do**
7: Update the membership matrix of each view $\mathbf{U}_k^{(t+1)}$ by using Eq. (21);
8: Update the transition matrix $\mathbf{P}_k^{(t+1)}$ by using Eq. (15);
9: Update the centroids in each view $\mathbf{C}_k^{(t+1)}$ by using Eq. (13);
10: Update the self-tuned weights $\{\omega_k^{(t+1)}\}_{k=1}^{v}$ by using Eq. (10);
11: **end for**
12: Update the centroids of latent features $\mathbf{O}^{(t+1)}$ by using Eq. (31);
13: Update the consensus matrix $\mathbf{F}^{(t+1)}$ by using Eq. (35);
14: Set $t = t + 1$;
15: **until** $\left\|\mathbf{F}^{(t)} - \mathbf{F}^{(t-1)}\right\|_{\infty} < \epsilon$ or $t \geq t_{max}$.
16: Determine the cluster assignment according to the maximum membership degrees in the consensus matrix $\mathbf{F}^{(t)}$.

fuzzy clustering with the latent information. The overall computational complexity of our approach is given in Table 4.

In the first phase, generating anchors takes $\mathcal{O}(nmQt_1)$, where t_1 represents the number of iterations required to perform the K-means algorithm, and $Q = \sum_{k=1}^{v} q_k$ denotes the total dimensions of features across different views. The construction of the cross-view anchor graph takes $\mathcal{O}(nrv)$, and the extraction of the latent information feature takes $\mathcal{O}(q_h(n+rv)^2)$ by using the SVD operation, where q_h is the dimensionality of latent features. Note that the extraction of latent feature matrix is executed only once. Therefore, the computational complexity of the first phase of our approach can be approximated as $\mathcal{O}(nmQt_1)$.

In each iteration of the second phase, it takes $\mathcal{O}(nmv)$, $\mathcal{O}(vm^2)$, $\mathcal{O}(nmQ)$, and $\mathcal{O}(nmQ)$ to update the self-tuned weights ω, transition matrices $\mathbf{P}$, centroids $\mathbf{C}$ of raw data in each view, membership matrices $\mathbf{U}$ in Eqs. (10), (15), (13), (21), respectively. The updating of the consensus matrix $\mathbf{F}$ in Eq. (35) takes $\mathcal{O}(nmQ)$. Additionally, updating the centroids $\mathbf{O}$ of latent features in Eq. (31) requires $\mathcal{O}(nmq_h)$. Therefore, the time complexity of the second phase is $\mathcal{O}((nmv + vm^2 + 3nmQ + nmq_h)t_2)$, where t_2 represents the iterations in this phase. Since the amount of data points is significantly greater than the number of clusters, anchors, and feature dimensions, i.e. $n \gg m, r, Q$, and q_h, the computational complexity of updating variables in the second phase can be approximated as linear complexity $\mathcal{O}(n)$.

3.4. Convergence analysis

We provide a proof for the convergence of the proposed algorithms using the following lemma.

Lemma 1 ([24]). *For any positive values a and b, the following inequality holds:*

$$a - \frac{a^2}{2b} \leq b - \frac{b^2}{2b} \tag{36}$$

Since the OMVFC-LICAG algorithm degenerates to the OMVFC when the hyperparameter β equals 0, we only provide the convergence analysis of the OMVFC-LICAG algorithm.

Theorem 1. *At each iteration of the OMVFC-LICAG algorithm, the objective function of problem (28) with respect to* $\mathbf{F}$ *will decrease monotonically until convergence is achieved, and the final optimization target* $\mathbf{F}$ *will converge to its local optimum.*

Proof of Theorem 1. Let $\mathbf{F}^{(t)}$ be the solution of $\mathbf{F}$ after t iterations and define the following equation:

$$f(\mathbf{F}) = \beta \operatorname{Tr}\left(\mathbf{D}_H^{\top}\mathbf{F}\right). \tag{37}$$

Then, the solution of $\mathbf{F}$ in the next iteration $t+1$ is obtained by

$$\mathbf{F}^{(t+1)} = \min_{\mathbf{F}} \sum_{k=1}^{v} \omega_k \left\|\mathbf{F} - \mathbf{P}_k\mathbf{U}_k\right\|_F^2 + f(\mathbf{F}). \tag{38}$$

Combining with the definition of ω_k in Eq. (10), the following inequality holds:

$$\sum_{k=1}^{v} \frac{\left\|\mathbf{F}^{(t+1)} - \mathbf{P}_k\mathbf{U}_k\right\|_F^2}{\left\|\mathbf{F}^{(t)} - \mathbf{P}_k\mathbf{U}_k\right\|_F} + f\left(\mathbf{F}^{(t+1)}\right) \leq \sum_{k=1}^{v} \frac{\left\|\mathbf{F}^{(t)} - \mathbf{P}_k\mathbf{U}_k\right\|_F^2}{\left\|\mathbf{F}^{(t)} - \mathbf{P}_k\mathbf{U}_k\right\|_F} + f\left(\mathbf{F}^{(t)}\right). \tag{39}$$

Based on Lemma 1, we have

$$\begin{aligned} &\sum_{k=1}^{v}\left(\left\|\mathbf{F}^{(t+1)} - \mathbf{P}_k\mathbf{U}_k\right\|_F - \frac{\left\|\mathbf{F}^{(t+1)} - \mathbf{P}_k\mathbf{U}_k\right\|_F^2}{\left\|\mathbf{F}^{(t)} - \mathbf{P}_k\mathbf{U}_k\right\|_F}\right) \\ &\leq \sum_{k=1}^{v}\left(\left\|\mathbf{F}^{(t)} - \mathbf{P}_k\mathbf{U}_k\right\|_F - \frac{\left\|\mathbf{F}^{(t)} - \mathbf{P}_k\mathbf{U}_k\right\|_F^2}{\left\|\mathbf{F}^{(t)} - \mathbf{P}_k\mathbf{U}_k\right\|_F}\right). \end{aligned} \tag{40}$$

By adding both sides in Eqs. (39) and (40), we have

$$\sum_{k=1}^{v} \left\|\mathbf{F}^{(t+1)} - \mathbf{P}_k\mathbf{U}_k\right\|_F + f\left(\mathbf{F}^{(t+1)}\right) \leq \sum_{k=1}^{v} \left\|\mathbf{F}^{(t)} - \mathbf{P}_k\mathbf{U}_k\right\|_F + f\left(\mathbf{F}^{(t)}\right), \tag{41}$$

which completes the proof. □

Similarly, the local optimum solutions of $\mathbf{P}$ and $\mathbf{U}$ can be obtained by Eqs. (15) and (21), respectively. The optimization problem (28) with respect to $\mathbf{C}$ and $\mathbf{O}$ are convex, and $\mathbf{C}$ and $\mathbf{O}$ will converge to their local optimum solutions by using Eqs. (13) and (31), respectively. It proves that the objective function (28) will monotonically decrease until convergence is achieved, and all variables in Eq. (28) will converge to their local optimum solutions.

4. Experiments and results

Extensive experiments are conducted in this work to comprehensively evaluate the effectiveness and efficiency of our algorithm on eight multi-view datasets. The experiments are carried out on a desktop computer with a 3.80 GHz Intel(R) Core(TM) i7-10700K CPU, 32 GB RAM, and MATLAB R2018b (64-bit).

4.1. Datasets and evaluation metrics

In the experimental evaluation, we assess the performance of competitive methods on eight widely-adopted multi-view datasets, which can be categorized based on their modality into two groups: text and image [6,22,31]. The former category includes Movies, WebKB, Reuters-1500, and Wikipedia datasets, while the latter category encompasses Caltech101-all, MNIST-10k, Animal, and NUS-WIDE-OBJ datasets. Detail information of the above datasets is provided in Table 2. To evaluate the clustering performance of all compared methods, we employ three commonly used cluster evaluation metrics: Purity, Normalized Mutual Information (NMI), and Adjusted Rand Index (ARI) [47]. It is important to note that superior clustering results are indicated by higher scores of these evaluation metrics.

Table 2
Details of the experimental multi-view datasets.

Datasets	Instances	Clusters	Features
Movies	617	17	1878, 1398
WebKB	1051	2	1840, 3000
Reuters-1500	1500	6	2153, 24 893, 34 279, 15 506, 11 547
Wikipedia	2866	10	128, 10
Caltech101-all	9144	102	48, 40, 254, 1984, 512, 928
MNIST-10k	10 000	10	30, 9, 30
Animal	11 673	20	2688, 2000, 2001, 2000
NUS-WIDE-OBJ	30 000	31	65, 226, 145, 74, 129

4.2. Comparison algorithms and parameter settings

To prove the superiority of our approach, we performed a comparative analysis against eight state-of-the-art baseline algorithms, as follows.

1. AWP [28]: A multi-view clustering based on the adaptively weighted Procrustes approach.
2. FPMVS-CAG [31]: An efficient multi-view subspace clustering algorithm combined with the consensus anchor guidance approach.
3. LMVSC [29]: An anchor graph-based multi-view subspace clustering for large-scale datasets.
4. SwMC [24]: A self-weighted multi-view clustering, which is a parameter-free algorithm.
5. UDBGL [13]: The multi-view clustering via a unified and discrete bipartite graph learning approach.
6. MCGC [26]: A multi-view consensus graph clustering algorithm.
7. MVSC [25]: A multi-view spectral clustering based on the bipartite graph.
8. MVASM [23]: The multi-view fuzzy clustering with adaptive sparse memberships and weight allocation algorithm.

To demonstrate the novelty of the proposed method, we provide Table 3 for show the differences among the competitive methods. As shown in Table 3, we put forward a novel cross-view bipartite graph construction method which fuses the latent information across different views. This method not only addresses the challenge of integrating data of diverse views but also enhances the computational efficiency of classical graph construction approaches. Furthermore, we incorporate a self-tuned weighting mechanism to dynamically assess the significance of each view, while effectively managing the interplay between exploring view-specified memberships and learning the consensus matrix. Finally, our approach is a one-step multi-view fuzzy clustering framework that eliminates the need for additional post-processing steps. Although the MVASM algorithm is also classified as a one-step framework, it solely relies on a single membership matrix to represent the affinity of data points and cluster centroids across all views. In contrast, our method explores the membership information of each view individually, and then derives the final membership matrix through a consensus matrix learning term. Consequently, our approach exhibits a more fine-grained nature and demonstrates superior performance. Additionally, we list the computational complexity and parameter settings of different methods in Table 4.

4.3. Clustering results

We perform all comparison methods on eight real-world datasets and present the scores of three evaluation metrics in Table 5. The table reports the best mean scores and standard deviations (Std) on 20 trials. Based on these results, we make the following four observations:

1. Our OMVFC-LICAG algorithm has demonstrated superior performance compared to state-of-the-art algorithms, achieving either the best or the second-best scores across all benchmark datasets in terms of the Purity, ARI, and NMI metrics. Notably, our algorithm outperforms all compared methods with respect to the Purity metric. Furthermore, our approach is only 0.66% lower than the best baseline algorithm (LMVSC) on the MNIST-10k dataset in terms of the NMI metric. Despite not achieving the top ranking on two out of eight datasets concerning the ARI metric, our proposed method still obtains the second-highest scores.
2. The OMVFC-LICAG algorithm outperforms the second-place methods, with improvements of at least 2.99%, 3.71%, and 4.13% in terms of the Purity, ARI, and NMI metrics, respectively. It demonstrates a significant enhancement in the clustering performance of our approach for multi-view clustering tasks when compared to existing methods.
3. Statistical analysis using the t-Test confirms that the proposed method is superior to or not significantly different from all baseline methods.
4. The performance evaluation of various clustering algorithms on the NUS-WIDE-OBJ dataset highlights the effectiveness of anchor graph-based approaches in handling large-scale datasets. In contrast, the spectral clustering-based method MVSC and classical graph-based methods (i.e. MCGC and SwMC) suffer from out-of-memory errors that limit their ability to process big data.

To further demonstrate the superiority of the proposed OMVFC-LICAG algorithm, we conducted a post-hoc test [48] on all comparison algorithms. For $N_{\text{method}} = 9$ compared methods, $N_{\text{dataset}} = 8$ datasets and the critical value $q_\alpha = 2.855$, the critical difference (CD) is computed by

$$CD = q_\alpha \sqrt{\frac{N_{\text{method}}\left(N_{\text{method}}+1\right)}{6N_{\text{dataset}}}} = 3.91. \tag{42}$$

As shown in Fig. 2, the OMVFC-LICAG algorithm achieves the highest average ranks among the comparative approaches in terms of all evaluation metrics. Furthermore, our approach exhibits statistically significant differences (beyond the CD value) compared to six out of eight baseline algorithms (AWP, MCGC, MVSC, MVASM, SwMC, and UDBGL). In summary, the proposed method obtains outstanding clustering results for multi-view data and is significantly superior to most comparison approaches.

4.4. Ablation study

To clarify the effectiveness of latent information extraction, one-step multi-view fuzzy clustering framework and self-tuned weighting mechanism, we conducted the ablation study on the proposed method. Specifically, we perform the K-means algorithm on the latent features to obtain the partition result, namely LICAG. The OMVFC is the proposed one-step multi-view fuzzy clustering algorithm, and the simplified version of the OMVFC that fixes the weights of all views to 1 is denoted as OMVFC*. Finally, the ultimate method that integrates all the above proposed strategies is referred to as the OMVFC-LICAG algorithm.

Fig. 3 presents box plots in terms of NMI scores for these four algorithms. We can find that the OMVFC algorithm outperforms the LICAG on the Movies, Wikipedia, Animal and NUS-WIDE-OBJ datasets, while the LICAG algorithm achieves better and more stable results than the OMVFC on the WebKB, MNIST-10k and Caltech101-all datasets. This observation is attributed to the fact that the OMVFC performs the view-specified membership matrix optimization and the consensus matrix learning simultaneously. When there is a substantial disparity in information across diverse views, and achieving a consensus matrix becomes challenging, the clustering performance of the OMVFC algorithm may be unstable. The latent information extracted from the cross-view anchor graph provides effective guidance in the early stages of optimization, resulting in the positive performance and stability of the OMVFC-LICAG algorithm.

Table 3
Differences among the compared methods.

Methods	Model	Graph construction method	Self-weighted	One-step
AWP	Spectral feature	KNN	Yes	Yes
FPMVS-CAG	Subspace clustering	Anchor graph	No	No
LMVSC	Subspace clustering	Anchor graph	No	No
SwMC	Graph-based clustering	Constrained Laplacian Rank	Yes	No
UDBGL	Subspace clustering	Bipartite graph	No	No
MCGC	Spectral feature	KNN	No	No
MVSC	Spectral clustering	Bipartite graph	No	No
MVASM	Fuzzy clustering	No graph information	No	Yes
Our	Fuzzy clustering	Cross-view bipartite graph	Yes	Yes

Table 4
The computational complexity and parameter settings of different methods.

Methods	Computational complexity	Parameter settings
AWP	$\mathcal{O}(nQ^2t_2)$	K nearest neighbors $\in [10, 50]$
FPMVS-CAG	$\mathcal{O}((Qm^2 + Qm^3 + nm^3)t_2)$	–
LMVSC	$\mathcal{O}(nrQt_1 + nr^2t_2)$	$r \in [5, 100]$, $\alpha \in [0.001, 100]$
SwMC	$\mathcal{O}(n^2 + (nv + mn^2)t_2)$	–
UDBGL	$\mathcal{O}(nQ + nQrt_1 + (nrm + nr^2(1+Q) + r^3)t_2)$	$r = m$, $\alpha \in [0.001, 1000]$, $\beta \in [0.001, 1000]$
MCGC	$\mathcal{O}(((t_1 + 1 + mv)n^2)t_2)$	$\beta \in [0.6, 100]$
MVSC	$\mathcal{O}(nrQt_1 + nr^2t_2)$	$r \in [max(m-20, 0), m+20]$, $\gamma \in [0.1, 100]$
MVASM	$\mathcal{O}(nvQ^2(M + 2\ m)t_2)$	$q \in [1, 6]$, $\gamma \in [0, 6]$
LICAG	$\mathcal{O}(nmQt_1)$	$q_h \in [max(m-20, 0), m+20]$, anchor rate $\in (0, 0.9]$
OMVFC	$\mathcal{O}(m(nv + vm + 3nQ)t_2)$	$\alpha \in [0.1, 10000]$, $\beta = 0$
OMVFC-LICAG	$\mathcal{O}(nmQt_1 + m(nv + vm + 3nQ + nq_h)t_2)$	$\alpha \in [0.1, 10000]$, $\beta \in [0.1, 10000]$

[1] t_1 and t_2 represent the iterations of the graph construction stage and the optimization stage, respectively.
[2] $Q = \sum_{k=1}^{v} q_k$ represents the total dimensionality of the features across all views, and q_h denotes the dimensionality of the latent features in our approach.
[3] n, m, r represent the numbers of data points, clusters, and anchors, respectively.
[4] M is the time complexity of the Newton's method in the MVASM algorithm.

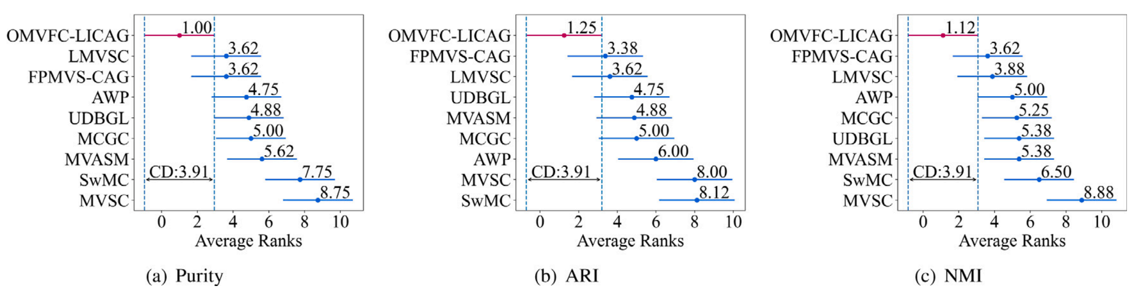

Fig. 2. The post-hoc test for nine competitive approaches, with dots indicating average ranks and bars representing the ranges of significant differences.

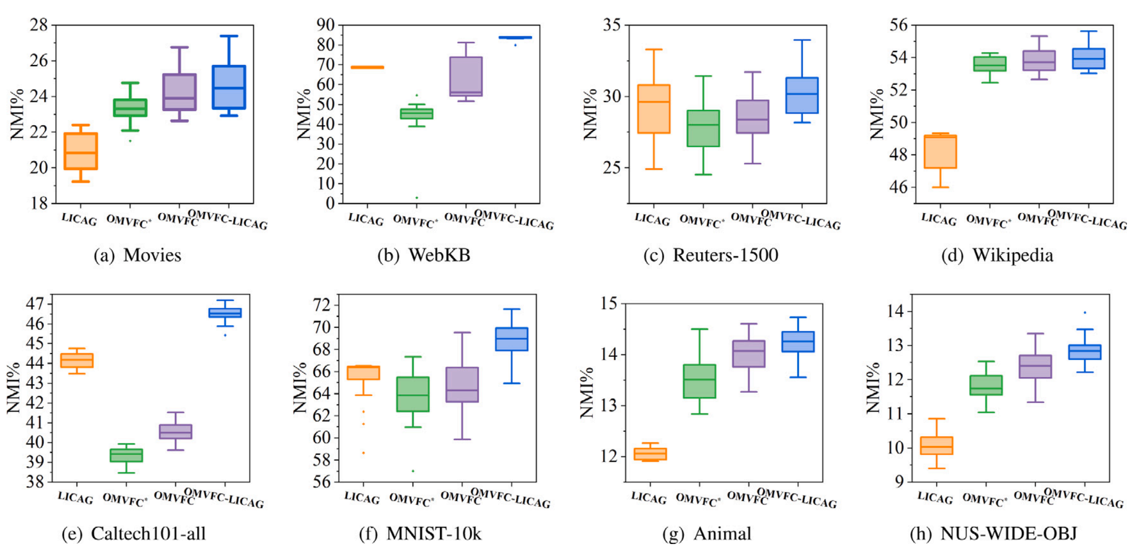

Fig. 3. Ablation study for the latent information extraction, one-step multi-view fuzzy clustering framework and self-tuned weighting mechanism.

Table 5
Clustering results of different algorithms on eight datasets.

Methods	AWP	FPMVS-CAG	LMVSC	MCGC	MVSC	MVASM	SwMC	UDBGL	Ours
Purity Mean% ± Std%									
Movies	21.56 ± 0.00	25.68 ± 1.01	23.18 ± 0.00	18.15 ± 0.00	11.86 ± 0.45	16.55 ± 3.05	14.51 ± 0.08	20.75 ± 0.00	**26.51 ± 1.31**
WebKB	83.35 ± 0.00	96.87 ± 0.17	95.15 ± 0.00	83.73 ± 0.00	79.20 ± 2.79	95.59 ± 1.46	79.84 ± 1.00	91.82 ± 0.00	**98.07 ± 0.16**
Reuters-1500	43.93 ± 0.00	57.87 ± 1.82	54.17 ± 0.00	49.61 ± 0.12	28.93 ± 1.23	52.44 ± 2.06	34.14 ± 0.33	44.20 ± 0.00	**58.38 ± 2.47**
Wikipedia	50.94 ± 0.00	33.40 ± 2.97	62.15 ± 0.00	57.33 ± 0.03	22.26 ± 0.51	47.71 ± 5.35	24.18 ± 3.56	53.91 ± 0.00	**62.71 ± 1.68**
Caltech101-all	43.47 ± 0.00	40.28 ± 0.19	37.80 ± 0.00	42.14 ± 0.51	20.63 ± 1.01	23.16 ± 0.28	29.22 ± 3.47	31.94 ± 0.00	**45.46 ± 0.55**
MNIST-10k	68.79 ± 0.00	71.07 ± 8.65	76.27 ± 0.00	75.33 ± 0.00	41.59 ± 6.86	72.43 ± 2.32	70.62 ± 0.25	65.33 ± 0.00	**77.83 ± 3.43**
Animal	19.07 ± 0.00	20.29 ± 0.43	18.55 ± 0.00	17.88 ± 0.05	13.72 ± 0.71	16.52 ± 0.45	11.57 ± 1.78	20.40 ± 0.00	**20.50 ± 0.59**
NUS-WIDE-OBJ	24.01 ± 0.00	23.59 ± 0.21	22.89 ± 0.00	OM	OM	21.73 ± 0.37	OM	23.61 ± 0.00	**24.62 ± 0.58**
t-Test(S/N/I)	8/0/0	6/2/0	6/2/0	8/0/0	8/0/0	8/0/0	8/0/0	7/1/0	
↑	7.37	5.63	2.99	8.74	24.49	8.49	18.75	7.76	
Adjusted Rand Index (ARI) Mean% ± Std%									
Movies	3.98 ± 0.00	**7.18 ± 1.57**	4.19 ± 0.00	1.66 ± 0.00	0.79 ± 0.12	3.68 ± 1.68	0.54 ± 0.02	2.95 ± 0.00	6.77 ± 1.37
WebKB	25.62 ± 0.00	87.21 ± 0.70	78.70 ± 0.00	27.41 ± 0.00	5.04 ± 13.83	81.87 ± 5.71	8.64 ± 5.02	67.93 ± 0.00	**89.51 ± 0.71**
Reuters-1500	3.86 ± 0.00	**26.88 ± 4.79**	21.22 ± 0.00	15.64 ± 0.16	0.25 ± 0.49	22.18 ± 4.40	0.01 ± 0.11	6.34 ± 0.00	25.96 ± 3.80
Wikipedia	25.19 ± 0.00	9.83 ± 2.35	43.91 ± 0.00	31.12 ± 0.04	10.10 ± 0.22	32.73 ± 6.11	0.25 ± 0.42	34.58 ± 0.00	**44.09 ± 2.31**
Caltech101-all	16.47 ± 0.00	21.40 ± 0.20	16.92 ± 0.00	18.58 ± 0.14	5.71 ± 1.79	5.55 ± 0.49	0.70 ± 0.31	7.67 ± 0.00	**21.84 ± 1.43**
MNIST-10k	52.75 ± 0.00	53.66 ± 9.33	57.83 ± 0.00	61.15 ± 0.00	22.34 ± 6.19	56.35 ± 3.65	57.33 ± 1.01	47.40 ± 0.00	**62.97 ± 3.15**
Animal	4.51 ± 0.00	5.16 ± 0.19	4.65 ± 0.00	5.12 ± 0.02	2.05 ± 0.60	5.11 ± 0.64	1.57 ± 0.50	5.19 ± 0.00	**5.63 ± 0.64**
NUS-WIDE-OBJ	4.51 ± 0.00	4.69 ± 0.00	4.78 ± 0.00	OM	OM	4.18 ± 0.35	OM	4.80 ± 0.00	**5.11 ± 0.63**
t-Test(S/N/I)	8/0/0	5/3/0	7/1/0	8/0/0	8/0/0	8/0/0	8/0/0	8/0/0	
↑	15.62	5.73	3.71	12.65	26.95	6.28	24.10	10.63	
Normalized Mutual Information (NMI) Mean% ± Std%									
Movies	17.03 ± 0.00	23.54 ± 0.86	20.86 ± 0.00	12.99 ± 0.00	8.20 ± 0.78	15.76 ± 2.75	17.98 ± 0.02	18.15 ± 0.00	**24.18 ± 1.00**
WebKB	25.86 ± 0.00	76.90 ± 0.89	67.62 ± 0.00	27.18 ± 0.00	4.93 ± 10.35	71.53 ± 6.98	11.73 ± 5.46	55.54 ± 0.00	**82.33 ± 1.00**
Reuters-1500	23.86 ± 0.00	30.22 ± 2.76	28.32 ± 0.00	20.67 ± 0.11	0.96 ± 0.38	22.78 ± 2.35	13.73 ± 0.36	13.33 ± 0.00	**31.36 ± 2.07**
Wikipedia	30.98 ± 0.00	14.29 ± 2.36	53.42 ± 0.00	44.99 ± 0.03	10.78 ± 0.23	47.77 ± 3.34	17.98 ± 3.77	44.93 ± 0.00	**54.11 ± 0.69**
Caltech101-all	45.79 ± 0.00	45.44 ± 0.13	40.38 ± 0.00	43.66 ± 0.41	20.49 ± 0.87	24.88 ± 0.49	36.28 ± 3.83	33.59 ± 0.00	**47.35 ± 0.49**
MNIST-10k	67.52 ± 0.00	63.04 ± 5.58	**70.11 ± 0.00**	68.59 ± 0.00	38.12 ± 5.41	67.63 ± 2.47	68.20 ± 1.76	58.41 ± 0.00	69.45 ± 1.91
Animal	12.65 ± 0.00	13.67 ± 0.64	10.88 ± 0.00	13.57 ± 0.00	6.59 ± 0.96	11.60 ± 0.88	11.21 ± 1.42	13.04 ± 0.00	**14.25 ± 0.28**
NUS-WIDE-OBJ	12.10 ± 0.00	12.47 ± 0.12	11.50 ± 0.00	OM	OM	10.37 ± 0.27	OM	12.59 ± 0.00	**13.10 ± 0.33**
t-Test(S/N/I)	8/0/0	7/1/0	7/1/0	7/1/0	8/0/0	8/0/0	8/0/0	8/0/0	
↑	12.54	7.07	4.13	13.06	30.76	7.98	19.88	10.82	

[1] The highest and second-highest scores are marked in boldface and underlined, respectively.
[2] The abbreviation "OM", stands for out of memory, and ↑ denotes the average improvement of our approach over the corresponding compared algorithm.
[3] In the t-Test, the 'S/N/I' notation denotes whether our algorithm is superior to, not significantly different from, or inferior to the baseline method.

In addition, we also evaluated the effectiveness of the self-tuned weighting mechanism in our proposed method by comparing the performance of our approach with (i.e. OMVFC) and without the weighting mechanism (i.e. OMVFC*). The results in Fig. 3 validate that the OMVFC algorithm outperforms the OMVFC* algorithm, which discards the self-tuned weighting mechanism, across all benchmark datasets. Fig. 4 shows the view weight values calculated by the self-tuned weighting mechanism at each iteration. During the early optimization stage, when the quality of the view-specified membership matrix is poor, the self-tuned weighting mechanism assigns small weight to each view. As the membership matrices are optimized, the weights of each view gradually increase, and the influence of the view-specified membership on the final consensus matrix becomes stronger. Overall, the self-tuned weighting mechanism finely adjusts the contribution of single-view information to the consensus matrix based on the quality of their respective membership information, which improves the clustering performance of our approach.

4.5. Parameter sensitivity analysis

The effectiveness of the latent information extraction (LICAG) heavily depends on two parameters: the latent feature dimensions q_h and the anchor rate. Fig. 5 illustrates the Purity scores obtained by the LICAG algorithm with varying q_h and anchor rates. For small-scale datasets, including Movies, WebKB, Reuters-1500, and Wikipedia, the anchor rate ranges from 0.15 to 0.9. On larger datasets, such as Caltech101-all, MNIST-10k, and Animal, the anchor rate is between 0.005 and 0.05, and varies from 0.0005 to 0.003 on the NUS-WIDE-OBJ dataset. We can observe that setting q_h to approximately the number of clusters m results in better quality of latent features on most datasets. Additionally, The proposed method performs favorably with an anchor rate of less than 0.6 on the WebKB, Reuters-1500, and Wikipedia datasets. Moreover, our approach outperforms existing methods while utilizing less than 0.05 of the data points as anchors on the Caltech101-all, MNIST-10k, Animal and NUS-WIDE-OBJ datasets. These results demonstrate the efficiency and effectiveness of the latent information extraction based on the cross-view anchor graph.

In the experiments for the OMVFC-LICAG algorithm, we set the latent feature dimensions q_h and anchor rate to their optimal values and investigate the sensitivity of the hyperparameters α and β. The values of α and β are selected from the set $\{0.1, 1, 10, 100, 1000, 10000\}$. When $\alpha > 0$ and $\beta = 0$, the OMVFC-LICAG algorithm is simplified to the OMVFC algorithm. Conversely, when $\beta > 0$ and $\alpha = 0$, the proposed approach is equivalent to performing fuzzy clustering on the latent features. Fig. 6 illustrates the Purity scores obtained by the OMVFC-LICAG algorithm with varying values of α and β. The experimental results indicate that our approach achieves favorable scores when $\alpha > 100$ and β varies between 1 and 100.

4.6. Empirical complexity study

Table 6 presents a detailed runtime analysis of all comparison algorithms on the eight datasets. Notably, classical graph-based algorithms, such as SwMC, MCGC, and MVSC, exhibit considerably longer execution times than anchor graph-based methods, including AWP, FPMVS-CAG, LMVSC, UDBGL, MVASM, and OMVFC-LICAG, when processing large-scale datasets (i.e. Caltech101-all, MNIST-10k, Animal, and NUS-WIDE-OBJ). Furthermore, the SwMC, MCGC, and MVSC algorithms

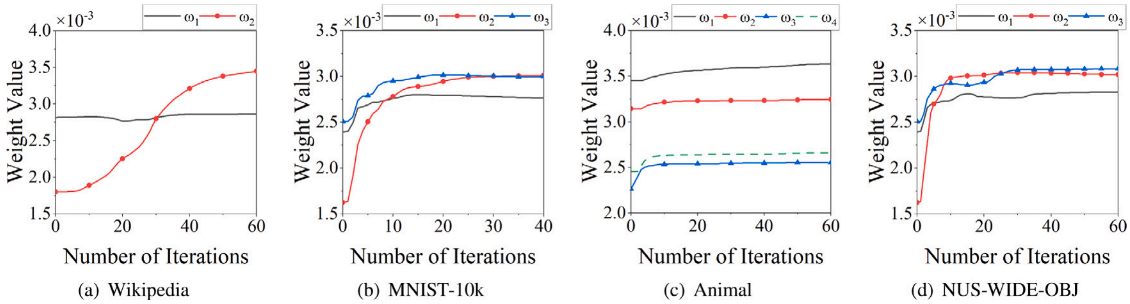

Fig. 4. The weight value curves on different datasets.

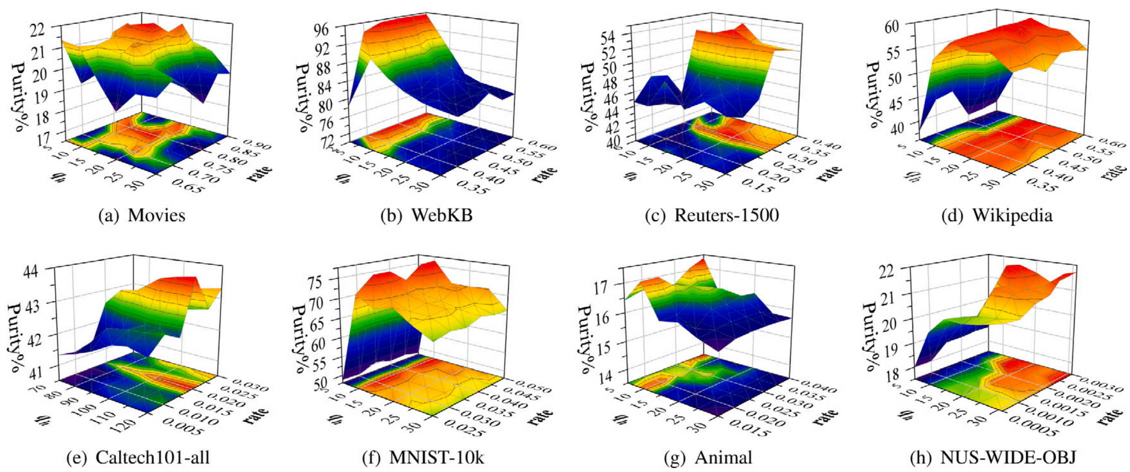

Fig. 5. The Purity scores achieved by the proposed method with various latent feature dimensions (q_h) and anchor rate.

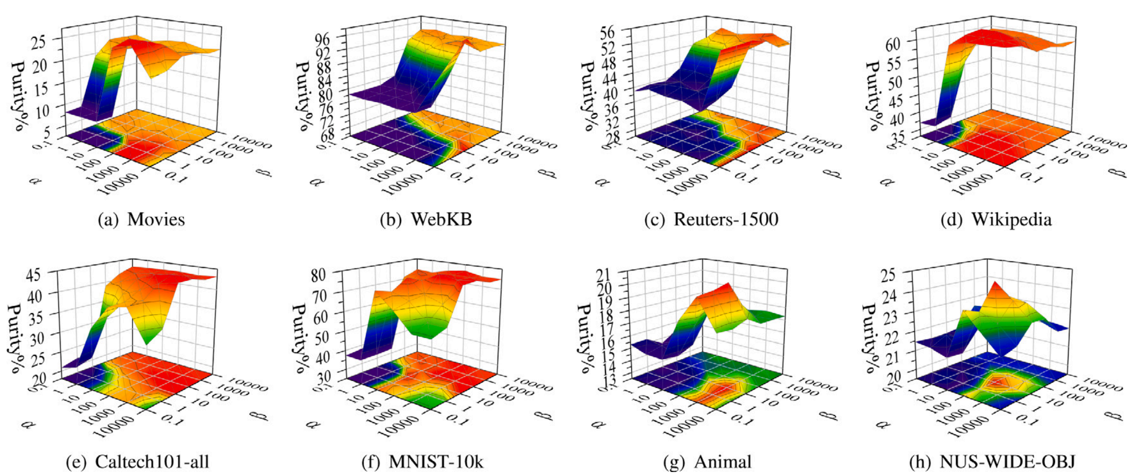

Fig. 6. The Purity scores achieved by the proposed method with various hyperparameters α and β.

encounter out-of-memory errors when processing the NUS-WIDE-OBJ dataset. Although the AWP, FPMVS-CAG, and MVASM algorithms have shorter execution times compared to the proposed method, our method yields significant improvements of at least 5.63%, 5.73% and 7.07% in terms of Purity, ARI and NMI metrics compared to these algorithms. In situations where the total dimensions of the data is high, the time consumption of our approach is mainly attributed to the latent information extraction (LICAG). The above observations demonstrate the importance of an efficient graph construction method for the graph-inspired multi-view clustering framework. Additionally, the time complexity of the proposed one-step multi-view fuzzy clustering algorithm (OMVFC) is linear, and its execution time is shorter than that of most competitive methods. Based on the experimental results of clustering performance and execution time, our ultimate method (OMVFC-LICAG) strikes a good balance between accuracy and efficiency, making it a promising approach for multi-view clustering tasks.

4.7. Empirical convergence study

Fig. 7 illustrates the convergence curves of the OMVFC-LICAG algorithm on eight multi-view datasets. The objective function values decrease rapidly in the first 20 iterations and become stable within 60 iterations on various datasets. These results validate that the proposed alternating optimization algorithm can ensure a good convergence of our model.

Table 6
The runtime results (second) of different methods on eight datasets.

Methods	Movies	WebKB	Reuters-1500	Wikipedia	Caltech101-all	MNIST-10k	Animal	NUS-WIDE-OBJ
AWP	0.01	0.01	0.01	0.02	1.67	0.04	0.74	2.17
FPMVS-CAG	16.88	17.78	577.08	15.27	72.43	21.47	129.9	52.32
LMVSC	11.97	12.3	26.49	12.88	39.34	21.44	135.15	109.33
SwMC	0.87	2.37	7.32	213.8	1813.47	11 801.1	1755.88	OM
UDBGL	12.58	12.71	83.43	30.58	108.9	39.67	242.36	622.07
MCGC	2.03	2.75	5.15	20.87	1289.4	261.05	1368.61	OM
MVSC	1.29	3.13	5.63	18.39	377.6	218.96	347.5	OM
MVASM	0.6	0.7	0.41	0.41	11.29	2.08	240	32.29
LICAG	1.65	3.05	84.32	13.42	32.30	17.10	176.01	41.19
OMVFC	0.94	0.68	4.76	3.96	71.13	16.15	58.2	114.9
OMVFC-LICAG	2.02	4.81	90.48	17.36	116.93	40.69	233.85	158.51

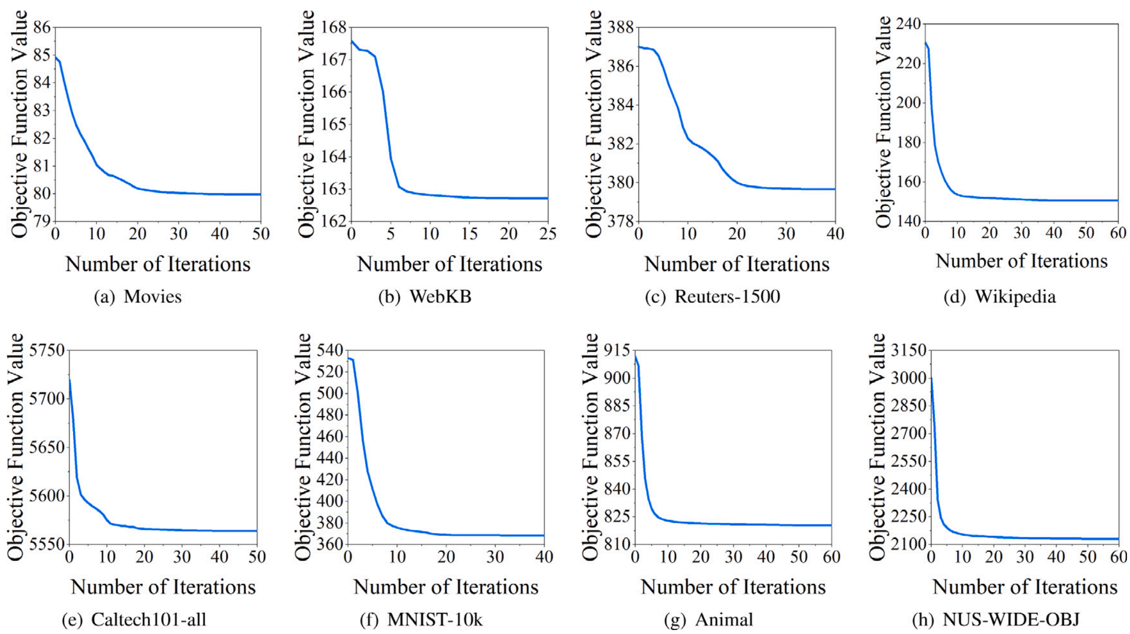

Fig. 7. The objective function curves on different datasets.

5. Conclusion

This paper put forward a novel multi-view fuzzy clustering algorithm, namely OMVFC-LICAG, which offers simplicity, scalability and self-tuning properties. The OMVFC-LICAG algorithm simultaneously implements view-specified membership exploration and consensus matrix learning on the multi-view data, allowing the final clustering result to be obtained in a straightforward and effective manner (simplicity). To further enhance the stability and accuracy of our framework, we exploit the latent information of raw data to guide the consensus matrix optimization. The latent information is extracted from the cross-view anchor graph, which is constructed by using a computationally efficient bipartite graph approach (scalability). Moreover, we introduce a self-tuned weighting mechanism to adaptively determine the importance of single-view information and prevent premature convergence of the consensus matrix to a local optimum (self-tuning). Experimental results of eight multi-view datasets verify the superiority of our approach.

Although the proposed method efficiently generates the consensus matrix by linearly combining view-specified membership matrices, its stability can be further improved. In the future, we plan to concatenate membership matrices across all views into a tensor and generate a better consensus matrix by imposing tensor-based regularization on the membership tensor, such as low-rank tensor [49] and tensor nuclear norm [50].

CRediT authorship contribution statement

Chuanbin Zhang: Methodology, Experiments, Validation, Writing – original draft. **Long Chen:** Methodology, Supervision, Writing – review & editing. **Zhaoyin Shi:** Writing – review & editing. **Weiping Ding:** Writing – review & editing.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Long Chen reports financial support was provided by Guangdong Basic and Applied Basic Research Foundation. Long Chen reports financial support was provided by Science and Technology Development Fund, Macao S.A.R. Long Chen reports financial support was provided by University of Macau and the University of Macau Development Foundation. Weiping Ding reports financial support was provided by National Natural Science Foundation of China. Weiping Ding reports financial support was provided by Natural Science Foundation of Jiangsu Province. Weiping Ding reports financial support was provided by Natural Science Key Foundation of Jiangsu Education Department.

Data availability

Data will be made available on request.

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References

[1] Q. Zheng, J. Zhu, Z. Li, Z. Tian, C. Li, Comprehensive multi-view representation learning, Inf. Fusion 89 (2023) 198–209.
[2] M. Shi, X. Zhao, X. Yin, X. Chang, F. Niu, J. Guo, Multiview latent structure learning: local structure-guided cross-view discriminant analysis, Knowl.-Based Syst. 276 (2023) 110707.
[3] A. Rahate, R. Walambe, S. Ramanna, K. Kotecha, Multimodal co-learning: challenges, applications with datasets, recent advances and future directions, Inf. Fusion 81 (2022) 203–239.
[4] W. Yan, M. Gu, J. Ren, G. Yue, Z. Liu, J. Xu, W. Lin, Collaborative structure and feature learning for multi-view clustering, Inf. Fusion 98 (2023) 101832.
[5] J. Zhao, X. Xie, X. Xu, S. Sun, Multi-view learning overview: recent progress and new challenges, Inf. Fusion 38 (2017) 43–54.
[6] X. Cai, D. Huang, G.-Y. Zhang, C.-D. Wang, Seeking commonness and inconsistencies: A jointly smoothed approach to multi-view subspace clustering, Inf. Fusion 91 (2023) 364–375.
[7] W. Yan, Y. Zhang, C. Lv, C. Tang, G. Yue, L. Liao, W. Lin, Gcfagg: global and cross-view feature aggregation for multi-view clustering, in: Proc. IEEE Conf. Comput. Vis. Pattern Recognit., 2023, pp. 19863–19872.
[8] H. Yang, Q. Gao, W. Xia, M. Yang, X. Gao, Multiview spectral clustering with bipartite graph, IEEE Trans. Image Process. 31 (2022) 3591–3605.
[9] Z. Li, C. Tang, X. Liu, X. Zheng, W. Zhang, E. Zhu, Consensus graph learning for multi-view clustering, IEEE Trans. Multimed. 24 (2022) 2461–2472.
[10] C. Tang, Z. Li, J. Wang, X. Liu, W. Zhang, E. Zhu, Unified one-step multi-view spectral clustering, IEEE Trans. Knowl. Data Eng. 35 (6) (2023) 6449–6460.
[11] S. El Hajjar, F. Dornaika, F. Abdallah, Multi-view spectral clustering via constrained nonnegative embedding, Inf. Fusion 78 (2022) 209–217.
[12] S. Shi, F. Nie, R. Wang, X. Li, Fast multi-view clustering via prototype graph, IEEE Trans. Knowl. Data Eng. 35 (1) (2023) 443–455.
[13] S.-G. Fang, D. Huang, X.-S. Cai, C.-D. Wang, C. He, Y. Tang, Efficient multi-view clustering via unified and discrete bipartite graph learning, IEEE Trans. Neural Netw. Learn. Syst. (2023) http://dx.doi.org/10.1109/TNNLS.2023.3261460, early access.
[14] M.-S. Chen, L. Huang, C.-D. Wang, D. Huang, J.-H. Lai, Relaxed multi-view clustering in latent embedding space, Inf. Fusion 68 (2021) 8–21.
[15] Z. Kang, G. Shi, S. Huang, W. Chen, X. Pu, J.T. Zhou, Z. Xu, Multi-graph fusion for multi-view spectral clustering, Knowl.-Based Syst. 189 (2020) 105102.
[16] T. Zhou, C. Zhang, X. Peng, H. Bhaskar, J. Yang, Dual shared-specific multiview subspace clustering, IEEE Trans. Cybern. 50 (8) (2020) 3517–3530.
[17] C. Zhang, H. Fu, Q. Hu, X. Cao, Y. Xie, D. Tao, D. Xu, Generalized latent multi-view subspace clustering, IEEE Trans. Pattern Anal. Mach. Intell. 42 (1) (2020) 86–99.
[18] B. Yang, X. Zhang, Z. Li, F. Nie, F. Wang, Efficient multi-view K-means clustering with multiple anchor graphs, IEEE Trans. Knowl. Data Eng. 35 (7) (2023) 6887–6900.
[19] C. Chen, Y. Wang, W. Hu, Z. Zheng, Robust multi-view k-means clustering with outlier removal, Knowl.-Based Syst. 210 (2020) 106518.
[20] L. Xing, H. Zhao, Z. Lin, B. Chen, Mixture correntropy based robust multi-view k-means clustering, Knowl.-Based Syst. 262 (2023) 110231.
[21] M.-S. Yang, K.P. Sinaga, Collaborative feature-weighted multi-view fuzzy c-means clustering, Pattern Recognit. 119 (2021) 108064.
[22] H. Wei, L. Chen, C.L.P. Chen, J. Duan, R. Han, L. Guo, Fuzzy clustering for multiview data by combining latent information, Appl. Soft Comput. 126 (2022) 109140.
[23] J. Han, J. Xu, F. Nie, X. Li, Multi-view K-means clustering with adaptive sparse memberships and weight allocation, IEEE Trans. Knowl. Data Eng. 34 (2) (2022) 816–827.
[24] F. Nie, J. Li, X. Li, Self-weighted multiview clustering with multiple graphs, in: Proc Int Jt. Conf Artif Intell, 2017, pp. 2564–2570.
[25] Y. Li, F. Nie, H. Huang, J. Huang, Large-scale multi-view spectral clustering via bipartite graph, in: Proc. AAAI Conf. Artif. Intell., Vol. 29, 2015, pp. 2750–2756.
[26] K. Zhan, F. Nie, J. Wang, Y. Yang, Multiview consensus graph clustering, IEEE Trans. Image Process. 28 (3) (2019) 1261–1270.
[27] F. Nie, W. Zhu, X. Li, Unsupervised large graph embedding, in: Proc. AAAI Conf. Artif. Intell., Vol. 31, 2017, pp. 2422–2428.
[28] F. Nie, L. Tian, X. Li, Multiview clustering via adaptively weighted procrustes, in: Proc. ACM Int. Conf. Knowl. Discov. Data Min., 2018, pp. 2022–2030.
[29] Z. Kang, W. Zhou, Z. Zhao, J. Shao, M. Han, Z. Xu, Large-scale multi-view subspace clustering in linear time, in: Proc. AAAI Conf. Artif. Intell., Vol. 34, 2020, pp. 4412–4419.
[30] S. Liu, X. Liu, S. Wang, X. Niu, E. Zhu, Fast incomplete multi-view clustering with view-independent anchors, IEEE Trans. Neural Netw. Learn. Syst. (2022) http://dx.doi.org/10.1109/TNNLS.2022.3220486, early access.
[31] S. Wang, X. Liu, X. Zhu, P. Zhang, Y. Zhang, F. Gao, E. Zhu, Fast parameter-free multi-view subspace clustering with consensus anchor guidance, IEEE Trans. Image Process. 31 (2021) 556–568.
[32] J. Wang, C. Tang, Z. Wan, W. Zhang, K. Sun, A.Y. Zomaya, Efficient and effective one-step multiview clustering, IEEE Trans. Neural Netw. Learn. Syst. (2023) http://dx.doi.org/10.1109/TNNLS.2023.3253246, early access.
[33] X. Cai, F. Nie, H. Huang, Multi-view k-means clustering on big data, in: Proc Int Jt. Conf Artif Intell, 2013, pp. 2598–2604.
[34] J. Xu, J. Han, F. Nie, X. Li, Re-weighted discriminatively embedded k -means for multi-view clustering, IEEE Trans. Image Process. 26 (6) (2017) 3016–3027.
[35] X. Liu, X. Zhu, M. Li, L. Wang, C. Tang, J. Yin, D. Shen, H. Wang, W. Gao, Late fusion incomplete multi-view clustering, IEEE Trans. Pattern Anal. Mach. Intell. 41 (10) (2019) 2410–2423.
[36] X. Liu, L. Liu, Q. Liao, S. Wang, Y. Zhang, W. Tu, C. Tang, J. Liu, E. Zhu, One pass late fusion multi-view clustering, in: Proc. Int. Conf. Mach. Learn., 2021, pp. 6850–6859.
[37] X. Hu, Y. Shen, W. Pedrycz, X. Wang, A. Gacek, B. Liu, Identification of fuzzy rule-based models with collaborative fuzzy clustering, IEEE Trans. Cybern. 52 (7) (2022) 6406–6419.
[38] Y. Jiang, F.-L. Chung, S. Wang, Z. Deng, J. Wang, P. Qian, Collaborative fuzzy clustering from multiple weighted views, IEEE Trans. Cybern. 45 (4) (2015) 688–701.
[39] Y. Wang, L. Chen, Multi-view fuzzy clustering with minimax optimization for effective clustering of data from multiple sources, Expert Syst. Appl. 72 (2017) 457–466.
[40] X. Zhu, S. Zhang, Y. Zhu, W. Zheng, Y. Yang, Self-weighted multi-view fuzzy clustering, ACM Trans. Knowl. Discov. Data 14 (4) (2020) 48:1–48:17.
[41] X. Liu, X. Zhu, M. Li, L. Wang, E. Zhu, T. Liu, M. Kloft, D. Shen, J. Yin, W. Gao, Multiple kernel kk-means with incomplete kernels, IEEE Trans. Pattern Anal. Mach. Intell. 42 (5) (2020) 1191–1204.
[42] Z. Shi, L. Chen, W. Ding, C. Zhang, Y. Wang, Parameter-free robust ensemble framework of fuzzy clustering, IEEE Trans. Fuzzy Syst. (2023) http://dx.doi.org/10.1109/TFUZZ.2023.3277692, early access.
[43] H. Wei, L. Chen, K. Ruan, L. Li, L. Chen, Low-rank tensor regularized fuzzy clustering for multiview data, IEEE Trans. Fuzzy Syst. 28 (12) (2020) 3087–3099.
[44] Y. Xie, W. Zhang, Y. Qu, L. Dai, D. Tao, Hyper-Laplacian regularized multilinear multiview self-representations for clustering and semisupervised learning, IEEE Trans. Cybern. 50 (2) (2020) 572–586.
[45] S. Huang, H. Wu, Y. Ren, I. Tsang, Z. Xu, W. Feng, J. Lv, Multi-view subspace clustering on topological manifold, in: Proc. Adv. Neural Inf. Process. Syst., Vol. 35, 2022, pp. 25883–25894.
[46] S. Wang, X. Liu, E. Zhu, C. Tang, J. Liu, J. Hu, J. Xia, J. Yin, Multi-view clustering via late fusion alignment maximization, in: Proc Int Jt. Conf Artif Intell, 2019, pp. 3778–3784.
[47] C. Zhang, L. Chen, Y.-P. Zhao, Y. Wang, C.L.P. Chen, Graph enhanced fuzzy clustering for categorical data using a Bayesian dissimilarity measure, IEEE Trans. Fuzzy Syst. 31 (3) (2023) 810–824.
[48] Y. Yao, Y. Li, B. Jiang, H. Chen, Multiple kernel K-means clustering by selecting representative kernels, IEEE Trans. Neural Netw. Learn. Syst. 32 (11) (2021) 4983–4996.
[49] M.-S. Chen, C.-D. Wang, J.-H. Lai, Low-rank tensor based proximity learning for multi-view clustering, IEEE Trans. Knowl. Data Eng. 35 (5) (2023) 5076–5090.
[50] C. Lu, Transforms based tensor robust PCA: corrupted low-rank tensors recovery via convex optimization, in: Proc. Int. Conf. Comput. Vis., 2021, pp. 1145–1152.